

9.1.2 The Problem of Robot Sensing

Much of what your senses “tell” you depends on what you already know about the kinds of things you might be sensing. What a particular sensation makes you think and conclude or do is a result of complex interaction between your sense organs and your brain. Robots “know” nothing, which is why sensing with physical sensors to solve these problems with sensor’s output and computers is very difficult.

What seems like a very simple sensing task becomes quite complex when applied to the robot. For example, for most people, it is pretty easy to tell if there is an apple on the table in front of them. You just open your eyes and look, immediately you know, “Yes, there it is”, or “No, there are no apples here.” You can also do it with your eyes closed by just reaching out and feeling or smelling for one. For a robot, it is not that simple.

Assume you have a robot with a light sensor that resembles a colour video camera. You would like to “teach” (program) your robot so that it can determine whether there is an apple in front of it. How will it be done? You may try like this—put something in front of the robot, and “tell” it is not an apple or it is an apple. How many things do you need to show? What about the apple size and shape variations?

Well, a good first step would be to let the robot understand what an apple is. How would you explain the apple to your robot?

The problem of robot sensing is not only difficult as illustrated above, it is greatly limited by the sensor capabilities. It is extremely difficult to build sensor systems that are as sensitive as human senses.

Properly designing or choosing and using sensors is an essential part of robotics, and sensors of all types are used in many different kinds of robots for different tasks. For example, some robots use only simple position sensors, while some may have a combination of special cameras and advanced software to gather information about their surroundings, and orient themselves based on the behaviour of other robots.

9.2 SENSORS IN ROBOTICS

Sensors are used in robotic systems for a variety of functions. The task planning and control algorithms, discussed in previous chapters, require, both on-line measurements of the parameters characterizing the internal state of the manipulator as well as its work environment. The most common and minimal use of sensors is to provide information about the status of links and joints of the manipulator and about the working environment of the robot. In addition, sensors may be deployed to provide data for inspection and quality control, safety monitoring, and to detect and resolve interlocks in the workcell.

The major functions of sensors in robots can be grouped into five basic categories.

1. Status sensors,
2. Environment sensors,
3. Quality control sensors,
4. Safety sensors, and
5. Workcell control sensors.

The role of the sensors in robotics under each of these categories is described below.

9.2.1 Status Sensors

The primary use of sensors on a robot is to sense position, velocity, acceleration, and/or torque/force at each joint of the manipulator for position and motion control. These sensors form an essential part of the basic or internal closed-loop control systems and are called *internal sensors* or *state* (or *status*) *sensors*. Internal sensors give feedback on the status of the manipulator itself. The degree of accuracy that can be achieved by a manipulator depends on the resolution and accuracy of internal sensors. Internal sensors must also be cost effective because they are required for each axis of the manipulator.

9.2.2 Environment Sensors

The second major function for sensors is to extract features of the objects in the workcell or surrounding environment of the robot. This knowledge is utilized by the computer controller to modify or adapt to a given situation. For example, if the robot has to process several types of different parts, each requiring a different sequence of actions by the robot, such as adjusting the gripper orientation or applying the exact gripping force, it must determine the required parameters for each part. Sensors for these functions are generally placed in the environment of the robot or are external to the manipulator and are called *external* or *environment sensors*. There are exceptions where an external sensor may be mounted on the manipulator such as a wrist force-torque sensor. Another example is a camera mounted on the wrist to “see” the part in workcell before gripping it.

External sensors would be used to perform one or more of the following functions:

- detect presence of workpiece.
- determination of position and/or orientation of workpiece and of objects to be articulated, or other objects present in the workcell.
- workpiece identification.
- determination of workpiece properties such as size, shape and so on.
- detect, identify obstacles in the environment, and provide information about their size, shape, location, speed and so on.
- provide information about the manipulator-environment interaction forces and torques.
- provide information about environmental variables such as temperature, humidity, and so on.

- determination of the position and orientation of the end-effector, joints, and links of the manipulator.

Sometimes, the accuracy requirements in a given application are more stringent than the inherent accuracy and repeatability of the manipulator. For example, the task of assembling two parts that have very small allowance and require close alignment that is smaller than the resolution of the manipulator. In such situations, the feedback from external sensors, such as vision, can be used to improve the accuracy of positioning of the robot.

All the information from the external sensors would have to be processed by the computer in real time to guide the manipulator in the execution of its programmed work cycle. An important sensing method is the vision system that might be employed to determine such characteristics as part location and orientation, size and shape, and many other properties. Robotic vision is discussed in detail in the later part of this chapter.

9.2.3 Quality Control Sensors

Inspection and quality control is an important function for robotic sensors. Traditionally, quality control is performed on statistical sampling basis, using manual inspection techniques. Because sensors can be used to determine a variety of part quality characteristics, the use of sensors permits 100 percent inspection. The external sensors used for environmental feedback can also be used for detecting faults and failures in the finished product. The inspection process can be made a part of the programmed work cycle and the sequence of operations performed by the manipulator may be linked to the result of inspection process. Computer vision, ultrasonic, sonic, and other sensors can be used for inspection. Vision system can provide a large variety of information about the work piece in addition to its position and orientation.

9.2.4 Safety Sensors

An important function of the sensors in robotics is safety and hazard monitoring. The safety of the workers and other equipment in the work environment, and that of the manipulator itself, are important concerns. For example, if there is power failure, all the links of the manipulator may fall to zero-gravity position instantaneously. This may injure the human beings in the vicinity or may damage the equipment or the manipulator itself. A solution is to sense the power failure and apply breaks to prevent the uncontrolled falling of links due to gravity.

9.2.5 Workcell Control Sensors

Another major use of sensor technology in robotics is to implement interlocks in workcell. An interlock in the work cycle is a situation that requires sequencing of tasks such that completion of a task must be ensured before proceeding to the next task. For example, a part must arrive on the conveyor before the gripper of the manipulator can pick it. This kind of verification of interlocked tasks can be done

by incorporating the signals from a variety of sensors in the robot program. Sensors may also be used to detect and resolve the interlocks in the workcell.

All the five categories of sensor functions described above require sensors to be an integral part of robot control system to accomplish specific control function. An intelligent robot requires an intelligent data acquisition system and multilevel hierarchical closed-loop control system. Figure 9.2 shows the multilevel control architecture of a computer-based intelligent robotic manipulator.

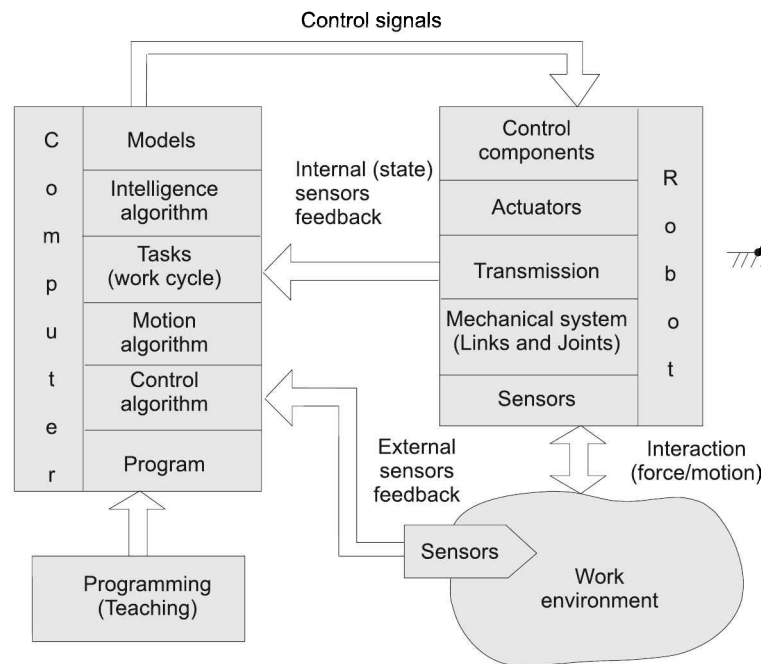


Fig. 9.2 Architecture of a computer-based intelligent robotic manipulator

Fusion of the available sensory data with task planning characterizes the robot as an intelligent robot, which is capable of perception of action. The control algorithms can guarantee the coordinated motion of the mechanical structure in correspondence to the task planning.

The signals from all the sensors on the manipulator and in the environment are fed to the computer-controller, which analyses them as per system models, task programs, control algorithms, motion algorithms, and intelligence algorithms and generates the control signals. These control signals are applied to the actuators to command and control the manipulator. The interaction of the manipulator with the work environment is sensed by the sensors and again fed back and the cycle goes on.

A broad classification of robotic sensors is discussed briefly in the next section.

9.2.6 Classification of Robotic Sensors

Robotic sensors can be classified in a number of broad groups in view of their applicability and use in robotics like property sensors, functional sensors, acoustic sensors, optic sensors and so on.

The schematic in Fig. 9.3 gives a possible classification hierarchy of the robotic sensors. The tasks and specific sensors in each group are also mentioned therein. These groupings are to some extent arbitrary and not unique. For example, external sensors could be subdivided into touch or tactile as well as into contact or noncontact sensors.

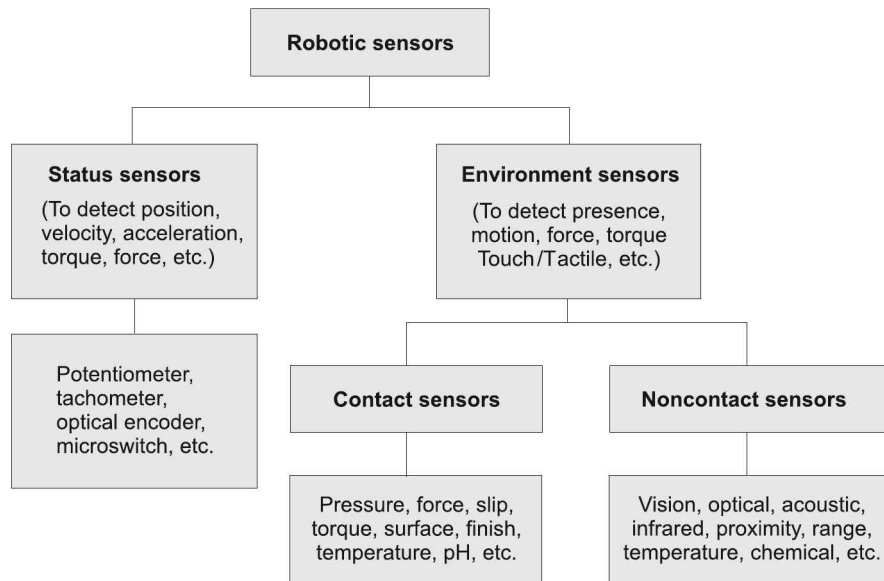


Fig. 9.3 A possible classification hierarchy of robotic sensors

Contact or touch sensors are one of the most common sensors in robotics. These are generally used to detect a change in position, velocity, acceleration, force, or torque at the manipulator joints and/or the end-effector. There are two main types, bumper and tactile. Bumper type detect whether they are touching anything, the information is either “yes” or “no”. They cannot give information about how hard is the contact or what they are touching. Tactile sensors are more complex and provide information on how hard the sensor is touched, or what is the direction and rate of relative movement.

The technologies used for sensors in this group are electric, electromagnetic, electronic, and optic. Some common robotic contact type sensors are potentiometers, tachometers, optical encoders, accelerometers, linear variable differential transformers (LVDT), and force-torque sensors.

The second important group is of the noncontact sensors, which are used to get parametric information about the environment, task, objects in the workspace and

to detect presence, distance, or features of the workpiece. The sensors in this group include proximity sensors, range sensors, temperature sensors, vision systems and so on. Various technologies such as optic, acoustic, infrared, electromagnetic and so on are used for sensors in this group. Proximity sensors are used in robotics to detect the presence of an object within a specified distance or find the accurate distance of the object. These are useful in object grasping or collision avoidance.

The vision system is probably the most powerful sensor as it can detect various features of the workpiece such as size, shape, geometry, distance, colour, texture, surface finish, and other attributes of the workpiece. It can also detect characteristics of the environment such as position and orientation of parts and obstacles, changes in these with time, presence of smoke, humidity, temperature gradient, and so on.

Some common kinds of sensors used in robotics are discussed here. The complete description of various sensor technologies and sensors and their detailed analysis can be found elsewhere.

9.3 KINDS OF SENSORS USED IN ROBOTICS

Many types of sensors are used in robotics. In this section, some common sensors suitable for robotic applications are discussed. These include touch and tactile sensors in the contact type and proximity, range and physical property sensors in the noncontact type.

Position sensing is the most fundamental need for robots to control position and orientation or motion of joints and links of the robot.

The measure of the position and rotation of the manipulator, that is, position of its end-effector or its joints is needed to get an accurate assessment of manipulator motions and positions. In many applications, there is a need to accurately know the position of the end-effector (tool or gripper) or to find the coordinate locations of objects in workspace, or to track motion to provide real-time adjustments. For all these, rotation sensors are very important in robotics to keep track of joint speeds and motions of end-effectors. There are very few devices to directly give absolute position for linear motions. Often linear motion is converted to rotary motion using rack and pinion drive and rotary sensors such as encoders are calibrated to give absolute linear position.

In industrial robotic manipulators currently manufactured, the revolute joints far outnumber the prismatic joints. Thus, sensing of rotation or angular position is the fundamental requirement, as stated above. Reliable position sensors are widely implemented in robots in the form of potentiometers, tachometers, encoders, resolvers, and accelerometers.

Proximity and touch sensors are generally used in intelligent grasping of work piece, while force-torque sensors provide feedback for object manipulation and control of its interaction with the environment. For example, these sensors are useful to prevent slipping or damage of the object once grasped or to apply the

right torque to tighten a nut. In contrast, the gross guidance information for a manipulator is provided by range and vision sensing. Vision sensing can also provide other useful information in addition to gross guidance information. The vision sensing fundamentals are discussed in later part of this chapter.

9.3.1 Acoustic Sensors

Acoustic or sonic sensors are commonly used for a wide variety of noncontact presence, proximity, distance measuring, or navigation applications. Sonic sensors can be simple or very complex. They typically transmit a short burst of ultrasonic sound towards the target, which reflects the sound back to the sensor. The system then counts the time for the echo to return to the sensor and calculates the distance of the target using the speed of sound in the medium. The wide variety of sensors currently available differ from one another acoustically. They operate at different frequencies and have different radiation patterns.

Ultrasonic sound is vibrations at a frequency above the range of human hearing, usually >20 kHz. Most ultrasonic sensors use a single transducer to both transmit the sound pulse and receive the reflected echo, typically operating at frequencies between 40 kHz and 250 kHz.

Silicon-based ultrasonic sensors and acoustic wave sensors are relatively new and extremely versatile microelectromechanical system (MEMS) sensors that are just beginning to realize their potential in robotics. They are competitively priced, inherently rugged, very sensitive, intrinsically reliable, and consistent in performance. These can be used for a large variety of applications, such as presence, proximity, velocity, acceleration, force torque, humidity, temperature, and so on.

Acoustic wave sensors send out acoustic waves, as the acoustic wave propagates through or on the surface of the material, changes in the characteristics of the propagation path affect the velocity and/or amplitude of the wave. These changes can be monitored by measuring the frequency or phase characteristics of the reflected wave. These changes can then be correlated to the corresponding physical quantity being measured. Array of ultrasonic sensors or acoustic wave sensors are useful for imaging and scene analysis. For a limited class of applications acoustical imaging may be complementary or even competitive to optical imaging.

9.3.2 Optic Sensors

Optic or light-based sensors are noncontact sensors and wide range of optic sensors based on different techniques are available for use in robotics. Optic sensors other than camera, used in vision systems, are discussed here.

Simple light sensors work with visible light and work on the principle of detecting the change in the intensity of light sent and received. The signal can be binary 0 (black or no light) or 1 (white or bright), or magnitude of light, say, between 0 and 100.

Optic sensors in nonvisible light include infrared, ultraviolet, and laser-beam sensors. An infrared sensor based tracking sensor system allows a full 3-D tracking within a space of few meters, with resolution of a fraction of a millimeter. Another eye safe laser based scanning sensor is used for navigation on automated guided vehicles, robots, and other material handling equipment.

9.3.3 Pneumatic Sensors

Applications of pneumatic sensors on robots are oldest and still convenient or suitable in some cases. Pneumatic sensors may be contact-less devices or contact sensors. They can be used as proximity sensors, touch sensors or pressure or force sensors.

Pneumatic proximity sensors are useful in workpiece grasping as they can provide accurate distance between fingers and workpiece, orientation of workpiece, finger closure signals and so on. Pressure-or vacuum-based sensors are very sensitive and are suitable for handling delicate and fragile objects. Counter-pressure based sensors are used in detecting or identifying the features such as edges, holes, shape, etc. and detecting the movement.

Some other conventional sensors that are used on robot are proximity and micro-switches, synchros, resolvers, differential transformers, eddy current sensors, hall-effect sensors, optical interrupters, tachometers and accelerometers.

9.3.4 Force/Torque Sensors

The most significant problem with force/torque sensor (FTS) is mounting it on the joint. For force/torque sensor to be sensitive, it must be flexible (not rigid). This makes the overall joint much more flexible contrary to rigidity required for precision and accuracy and causes a loss of controllability of the joint.

It is possible to estimate the joint torques and inertias without the use of force/torque sensors. Estimation algorithms based on actuator power input and inertia estimation are available. These methods are complex and very sensitive to propagated errors.

Apart from the joint forces/torques, the interaction between robot (its end-effector) and environment generates forces/torques that must be controlled to preserve the integrity of task being performed. For measuring these interaction forces, sensors are mounted between the wrist and the end-effector and are known as *Wrist Force Torque Sensors* (WFTS).

A strain-gauge based WFTS is shown in Fig. 9.4. The sensor has two annular rings, outer ring and inner ring, which are connected to wrist end and the end-effector end, respectively. These rings are connected to each other by a cross and on the arms of this cross, eight pairs of strain gauges are mounted as shown. By suitably selecting and connecting the gauges in the Wheatstone bridge the sensor detects three force components along the three principal axes and three moments along the three axes.

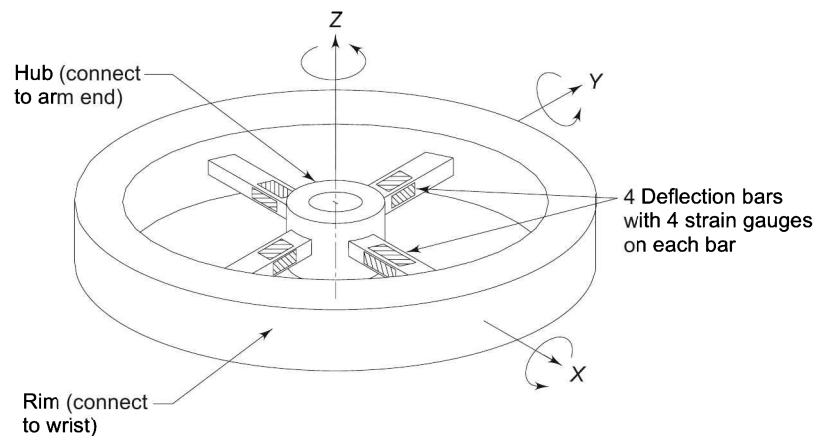


Fig. 9.4 A six-component strain-gauge based wrist force-torque sensor

Many other alternate designs are available for WFTS based not only on strain gauges but also on sonic and optic sensors. It is important to note that WFTS should not affect the positioning accuracy of the manipulator and, therefore, it must have high stiffness. Some other requirements of WFTS are: it must be small, light in weight, compact in design, sensitive, linear, and with low internal friction.

9.3.5 Optical Encoders

For position sensing in robotics, optical encoders are frequently used sensors because of their simple construction, low cost, ease of application and versatility. Optical encoder converts linear or angular displacement into digital code or pulse signals. Optical encoders are of two types:

Absolute encoders: They provide actual position relative to a fixed reference (zero) position. Their output is a digitally coded signal with distinct digital code indicative of each particular least significant increment of resolution.

Incremental encoders: These sense the position from previous position. Their output is a pulse for each increment of resolution but these make no distinction between increments.

The optical encoder is made up of three basic component: a light source, a rotary (or translatory) disc and a sensor, as illustrated in Fig. 9.5. The light source can be an incandescent lamp or an infrared light-emitting diode (LED). The disc has alternate opaque and transparent sectors, which are etched by means of a photographic process on a plastic disc or slots are cut on a metal disc. The sensor is a photodiode or a phototransistor, which senses the light (or no light) passing through sectors or slots of the disc. Finally, a conditioning electronics detects the light and dark signals and converts these signals into usable form of pulses or digital code.

The two types of optical encoders, incremental and absolute are described in detail in the following paragraphs.

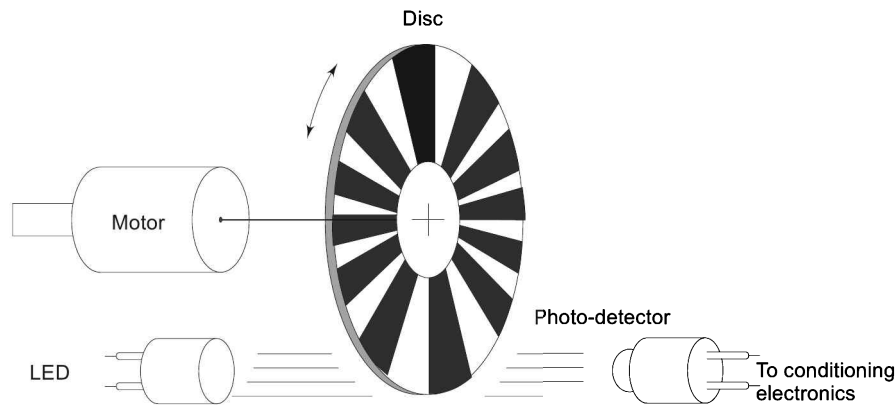


Fig. 9.5 Components of an optical shaft encoder

Incremental Optical Encoder The incremental optical encoder disc is connected to the motor shaft and rotates with motor. The rotation of the disc permits the light to reach the sensor, whenever the transparent slot comes in front of LED, thereby generating a series of pulses as output of sensor. These pulses are fed to a counter, which counts the number of pulses; the count being a measure of angle (or distance) through which the shaft has moved. By sampling the counter at regular intervals by means of clock pulses, the speed of rotation of the shaft can be obtained. The resolution of an incremental encoder is given by:

Basic resolution = $(360^\circ/m)$ where m = No. of sectors on disc, each sector is half transparent and half opaque. A typical incremental encoder having a disc with 2500 sectors (or lines) gives a basic resolution of 0.144° . Using another disk as a mask, using two or four optoelectronic channels or using multiplier circuits increases the accuracy and resolution of the incremental optical encoder. For example, if the pulses are increased by a factor of 4, by use of any of these approaches, the above incremental encoder will give a resolution of 0.036° .

Absolute Encoder Absolute optical encoder consists of a multiple track light source, a multi-track receiver and a multi-track rotary disk. The basic construction of absolute encoder is shown in Fig. 9.6(a) for the measurement of angular displacement. The absolute encoder disc has concentric circles of slots also known as tracks or channels and radial sectors to generate the pulses. As many LED-photodiode pairs as the number of tracks are needed. These may be suitably spread round the tracks to avoid signal interference.

The transparent and opaque slots in the tracks are arranged in such a way that the sequential output from the encoder is a number in binary code. The transparency is regarded as '1' and opaqueness as '0'. The tracks and slots of a 3-bit absolute encoder disc are illustrated in Fig. 9.6(b).

The number of bits in the binary code will be equal to the number of tracks on the disc. Thus, for an absolute encoder with a disc of 10 tracks, there will be 10 bits and the number of positions that can be detected is 2^{10} or 1024. This encoder will have a resolution of $360^\circ/1024 = 0.35^\circ$. An absolute encoder is a digital

transducer in the true sense as its output does not require any conditioning but can be directly read to get the absolute position information.

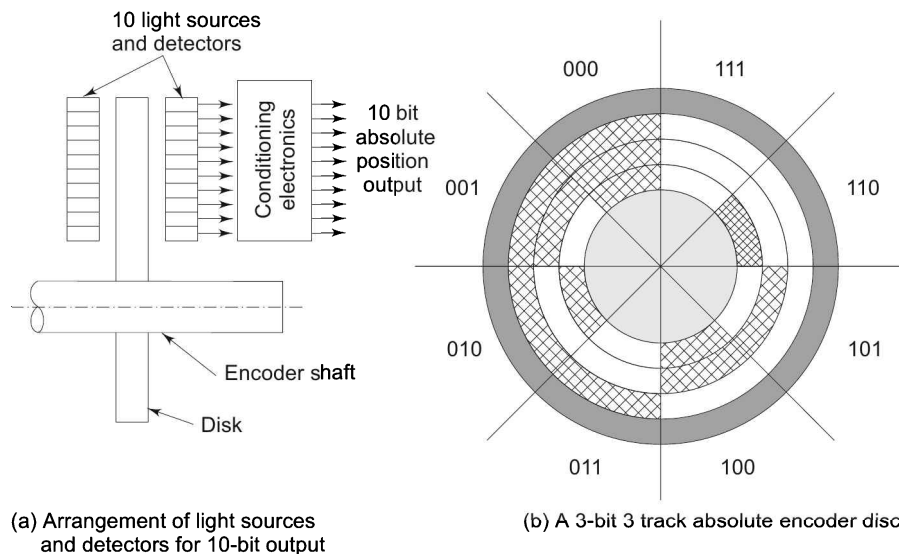


Fig. 9.6 Absolute optical encoder and its disc.

The chief disadvantage of optical encoders is that the output is volatile when power is off. There are ways to overcome this.

9.3.6 Choosing the Right Sensor

To decide what type of sensor or which sensor would be best for a particular situation, following guidelines may be used.

- Carefully analyze the nature of information needed.
- Find what information will be provided by the sensor.
- Determine how the sensor signals will be used.
- For the variable to be sensed determine: its nominal value, range of values, accuracy required, required reliability and speed of measurement etc.
- Estimate the environmental variations under which the sensing is to be done.
- Examine the nature of task dependent on the sensed signals, for example, critical tasks require reliable sensing.

Based on the above guidelines all possible sensors that can do the specified task are identified taking into account such other factors as sensitivity, linearity, in the desired range, maintainability, life, power consumption, ruggedness, availability and cost. Remember that there is always more than one way to sense the desired quantity and more than one type of sensor can do the job, hence, out of the several sensors available to get the desired information, one that is 'optimum' should be deployed.

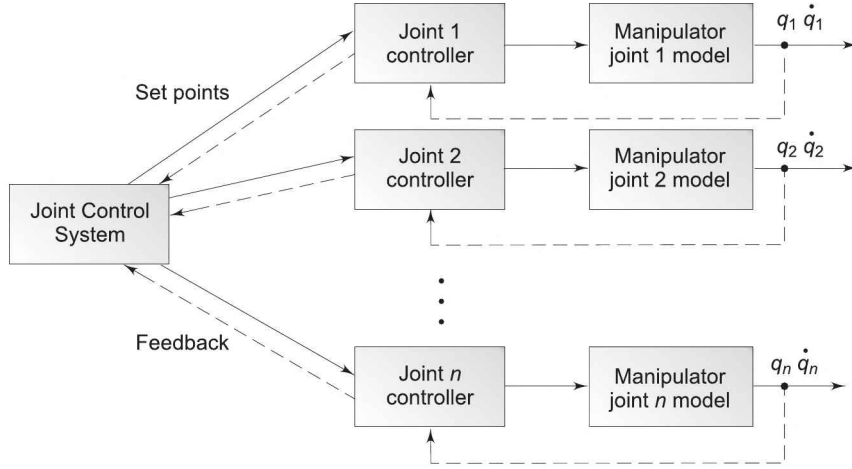


Fig. 8.2 Robot control architecture for an n -DOF manipulator

8.1 OPEN- AND CLOSE-LOOP CONTROL

In the case of *open-loop control* (no feedback), the actuator torque can be directly computed from the dynamic model of the manipulator. As seen in Chapter 6, Eq. (6.45), the dynamic behaviour of joint i of the manipulator is given by

$$\tau_i = \sum_j M_{ij} \ddot{q}_j + \sum_j \sum_k h_{ijk} \dot{q}_j \dot{q}_k + G_i \quad (8.1)$$

In this model, accurate computation of the parameters M_{ij} , h_{ijk} , and G_i is difficult. Moreover, this model does not include effects of friction, backlash and so on, at the joints and other external disturbances or noise. These effects are difficult to model and are highly nonlinear. Because of all these problems, the open-loop control has limited trajectory-tracking capabilities and is used in very few nonprecision applications.

The limitations of parametric inaccuracies and unmodelled nonlinear effects are overcome by using a *closed-loop control* scheme. Here, at every instant of time, the actual joint positions and velocities are measured directly by sensors such as encoders and tachometers mounted at the joints. These are used to compute the error between the desired and actual positions and velocities. A similar measurement and feedback is used for error in the desired end-effector position and/or force/torque. A control law then computes the joint torques required as a function of these servo errors using the dynamic model.

The interactive effects of inertial, Coriolis, centrifugal, and gravitational forces vary continuously with time. The computation of these from the dynamic model is complex and time consuming. This makes the actual real-time control extremely difficult.

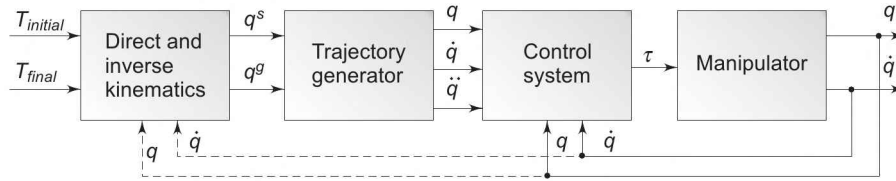


Fig. 8.3 Operations for a point-to-point motion control of a manipulator

The schematic description of the point-to-point motion control of an n -DOF manipulator arm is shown in Fig. 8.3. For this operation, the initial and final locations of the end-effector in Cartesian space, specified by the transformation matrices $T_{initial}$ and T_{final} , serve as the input. The inverse kinematics model computes the desired end-effector locations in joint space. Then, a trajectory generator computes the joint-position time histories, based on the joint-space algorithms discussed in Chapter 7. Depending on the servo error computed from the base reference values and the sensor measurements, the control system commands the individual actuators to achieve the desired motion.

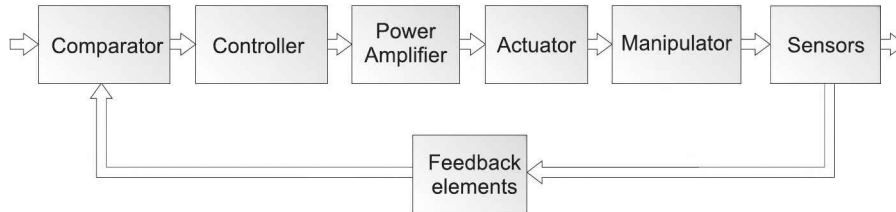


Fig. 8.4 General block diagram for manipulator control system

The block diagram in Fig. 8.4 illustrates the components of the control system to control the manipulator joints, consisting of actuators and the mechanical kinematics chain. A typical control system has a comparator (a small computer); a digital to analog converter (DAC); a controller to implement the control strategy; an amplifier to boost and transform the small command signals so that it can actuate the joint actuator; a sensor to detect the position and speed of joint and an analog to digital converter (ADC) in the feedback path.

8.2 THE MANIPULATOR CONTROL PROBLEM

The task to be performed by a manipulator may require either the execution of specific motions of the end-effector in free space and/or the application of specific contact forces at the end-effector, if the end-effector is constrained by the environment. For a given manipulator, many techniques can be used to control it to perform in the desired manner. The performance and range of applications of a manipulator is significantly influenced by the control scheme followed, as well as the way it is implemented. The mechanical design and configuration design also



Sensors in Robotics

Introduction

In Chap. 3, various sensor devices used as components of the robot control system were discussed. These devices included position sensors such as resolvers and encoders, and velocity sensors such as tachometers. Sensors can also be used as peripheral devices for the robot, just as end effectors discussed in Chap. 5 are used as peripheral devices. Most industrial applications require that sensors be employed in this way in order for the robot to operate with other pieces of equipment in the workcell.

In this chapter, the various types of sensors that are used as peripheral devices in robotics and the reasons why they are needed have been considered. We begin with a general discussion of transducers and sensors, then examine the sensors that are most commonly employed in robot applications. The final section in the chapter describes the four major categories of uses of sensor systems in robotics.

6.1 TRANSDUCERS AND SENSORS

A transducer is a device that converts one type of physical variable (e.g., force, pressure, temperature, velocity, flow rate, etc.) into another form. A common conversion is to electrical voltage, and the reason for making the conversion is that the converted signal is more convenient to use and evaluate using a digital computer. A sensor is a transducer that is used to make a measurement of a physical variable of interest. Some of the common sensors and transducers include strain gauges (used to measure force and pressure), thermocouples (temperatures), speedometers (velocity), and Pitot tubes (flow rates).

Any sensor or transducer requires calibration in order to be useful as a measuring device. Calibration is the procedure by which the relationship between the measured variable and the converted output signal is established.

Earlier we have seen that sensors can be clarified into external and internal sensors based on their functions.

Transducers and sensors can also be classified into two basic types depending on the form of the converted signal. The two types are:

1. Analog transducers
2. Digital transducers

Analog transducers provide a continuous analog signal such as electrical voltage or current. This signal can then be interpreted as the value of the physical variable that is being measured. Digital transducers produce a digital output signal, either in the form of a set of parallel status bits or a series of pulses that can be counted. In either form, the digital signal represents the value of the measured variable. Digital transducers are becoming more popular because of the ease with which they can be read as separate measuring instruments. In addition, they offer the advantage in automation and process control that they are generally more compatible with the digital computer than analog-based sensors.

In order to be useful as measuring devices, in robotics and in other applications, sensors must possess certain features. Some of the desirable engineering features of sensors and transducers are presented in Table 6-1. Few if any sensors have all of these desirable features, and a compromise must be made among them to select the best sensor for a given application.

Table 6.1 *Desirable features of sensors.*

1.	<i>Accuracy:</i> The accuracy of the measurement should be as high as possible. Accuracy is interpreted to mean that the true value of the variable can be sensed with no systematic positive or negative errors in the measurement. Over many measurements of the variable, the average error between the actual value and the sensed value will tend to be zero.
2.	<i>Precision:</i> The precision of the measurement should be as high as possible. Precision means that there is little or no random variability in the measured variable. The dispersion in the values of a series of measurements will be minimized.
3.	<i>Operating range:</i> The sensor should possess a wide operating range and should be accurate and precise over the entire range.
4.	<i>Speed of response:</i> The transducer should be capable of responding to changes in the sensed variable in minimum time. Ideally, the response would be instantaneous.
5.	<i>Calibration:</i> The sensor should be easy to calibrate. The time and trouble required to accomplish the calibration procedure should be minimum. Further, the sensor should not require frequent recalibration. The term 'drift' is commonly applied to denote the gradual loss in accuracy of the sensor with time and use, and which would necessitate recalibration.
6.	<i>Reliability:</i> The sensor should possess a high reliability. It should not be subject to frequent failures during operation.
7.	<i>Cost and ease of operation:</i> The cost to purchase, install, and operate the sensor should be as low as possible. Further, the ideal circumstance would be that the installation and operation of the device would not require a specially trained, highly skilled operator.

6.2 SENSORS IN ROBOTICS

The sensors used in robotics mainly for interaction with the environment include a wide range of devices which can be divided into the following general categories:

1. Tactile sensors
2. Proximity and range sensors
3. Miscellaneous sensors and sensor-based systems
4. Machine vision systems

We will discuss the first three of these categories in the sections that follow. The fourth category, machine vision, is examined in the next chapter. Table 6.2 provides a listing of some of the common sensors that are applicable in robotic workcells.

Table 6.2 *Sensor devices used in robot workcells.*

<i>Ammeter</i> —(miscellaneous) Electrical meter used to measure electrical current.
<i>Eddy current detectors</i> —(proximity sensor) Device that emits an alternating magnetic field at the tip of a probe, which induces eddy currents in any conductive object in the range of the device. Can be used to indicate presence or absence of a conductive object.
<i>Electrical contact switch</i> —(touch sensor) Device in which an electrical potential is established between two objects, and when the potential becomes zero, this indicates contact between the two objects. Not a commercial device. Can be used to indicate presence or absence of a conductive object.
<i>Infrared sensor</i> —(proximity sensor) Transducer which measures temperatures by the infrared light emitted from the surface of an object. Can be used to indicate presence or absence of a hot object.
<i>Limit switch</i> —(touch sensor) Electrical on–off switch actuated by depressing a mechanical lever or button on the device. Can be used to measure presence or absence of an object.
<i>Linear variable differential transformer</i> —(miscellaneous) Electromechanical transducer used to measure linear or angular displacement.
<i>Microswitch</i> —(touch sensor) Small electrical limit switch (see, limit switch). Can be used to indicate presence or absence of an object.
<i>Ohmmeter</i> —(miscellaneous) Meter used to measure electrical resistance.
<i>Optical pyrometer</i> —(proximity sensor, miscellaneous) Device used to measure high temperatures by sensing the brightness of an object's surface. Can be used to indicate presence or absence of a hot object.
<i>Photometric sensors</i> —(proximity sensor, miscellaneous) Various transducers used to sense light. Category includes photocells, photoelectric transducers, phototubes, photodiodes, phototransistors, and photoconductors. Can be used to indicate presence or absence of an object.
<i>Piezoelectric accelerometer</i> —(miscellaneous) Sensor used to indicate or measure vibration.
<i>Potentiometer</i> —(miscellaneous) Electrical meter used to measure voltage.
<i>Pressure transducers</i> —(miscellaneous) Various transducers used to indicate air pressure and other fluid pressures.

Contd.

<i>Radiation pyromete</i> —(proximity sensor, miscellaneous) Device used to measure high temperatures by sensing the thermal radiation emitting from the surface of an object. Can be used to indicate presence or absence of a hot object.
<i>Strain gage</i> —(force sensor) Common transducer used to measure force, torque, pressure, and other related variables. Can be used to indicate force applied to grasp an object.
<i>Thermistor</i> —(miscellaneous) Device based on electrical resistance used to measure temperatures.
<i>Thermocouple</i> —(miscellaneous) Commonly used device used to measure temperatures. Based on the physical principle that a junction of two dissimilar metals will emit an emf which can be related to temperature.
<i>Vacuum switches</i> —(proximity sensor, miscellaneous) Device used to indicate negative air pressures. Can be used with a vacuum gripper to indicate presence or absence of an object.
<i>Vision sensors</i> —(vision system) Advanced sensor system used in conjunction with pattern recognition and other techniques to view and interpret events occurring in the robot workplace.
<i>Voice sensors</i> —(voice and speed recognition) Advanced sensor system used to communicate commands or information orally to the robot.

6.3 TACTILE SENSORS

Tactile sensors are devices which indicate contact between themselves and some other solid object. Tactile sensing devices can be divided into two classes: touch sensors and force sensors. Touch sensors provide a binary output signal which indicates whether or not contact has been made with the object. Force sensors (also sometimes called stress sensors) indicate not only that contact has been made with the object but also the magnitude of the contact force between the two objects.

6.3.1 Touch Sensors

Touch sensors are used to indicate that contact has been made between two objects without regard to the magnitude of the contacting force. Included within this category are simple devices such as limit switches, microswitches, and the like. The simpler devices are frequently used in the design of interlock systems in robotics. For example, they can be used to indicate the presence or absence of parts in a fixture or at the pick up point along a conveyor. Another use for a touch-sensing device would be as part of an inspection probe which is manipulated by the robot to measure dimensions on a workpart. A robot with six degrees of freedom would be capable of accessing surfaces on the part that would be difficult for a three-axis coordinate measuring machine, the inspection system normally considered for such an inspection task. Unfortunately, the robot's accuracy would be a limiting factor in contact inspection work.

6.3.2 Force Sensors

The capacity to measure forces permits the robot to perform a number of tasks. These include the capability to grasp parts of different sizes in material handling, machine loading, and assembly work, applying the appropriate level of force for the given part. In assembly applications, force sensing could be used to determine if the screws have become cross-threaded or if the parts are jammed.

Force sensing in robotics can be accomplished in several ways. A commonly used technique is a 'force-sensing wrist.' This consists of a special load-cell mounted between the gripper and the wrist. Another technique is to measure the torque being exerted by each joint. This is usually accomplished by sensing motor current for each of the joint motors. Finally, a third technique is to form an array of force-sensing elements so that the shape and other information about the contact surface can be determined. We discuss these three possibilities in the paragraphs that follow.

Force-sensing wrist The purpose of a force-sensing wrist is to provide information about the three components of force (F_x , F_y , and F_z) and the three moments (M_x , M_y , and M_z) being applied at the end effector. One possible construction of a force-sensing wrist is illustrated in Fig. 6.1. The device consists of a metal bracket fastened to a rigid frame. The frame is mounted to the wrist of the robot and the tool is mounted to the center of the bracket. The figure shows how the sensors might react to a moment applied to the bracket due to forces and moments on the tool.

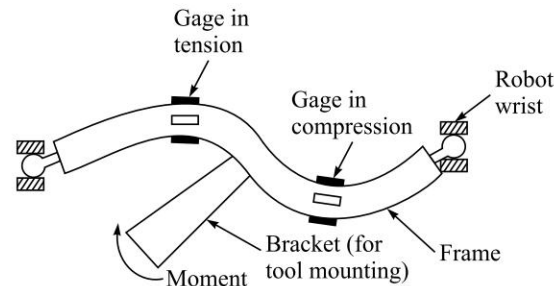


Fig. 6.1 Possible configuration of sensing device used for a force sensing wrist, showing deflection (exaggerated) due to a moment about one of the axes.

Since the forces are usually applied to the wrist in combinations it is necessary to first resolve the forces and moments into their six components. This kind of computation can be carried out by the robot controller (if it has the required computational capability) or by a specialized amplifier designed for this purpose. Based on these calculations, the robot controller can obtain the required information of the forces and moments being applied at the wrist. This information could be used for a number of applications. As an example, an insertion operation (e.g., inserting a peg into a hole in an assembly application) requires that there are no side forces being applied to the peg. Another example is where the robot's end effector is required to follow along an edge or contour of an irregular surface. This is called force accommodation. With this technique, certain forces are set equal to zero while others are set equal to specific values. Using force accommodation, one could command the robot to follow the edge or contour by maintaining a fixed velocity in one direction and fixed forces in other directions.

The robot equipped with a force-sensing wrist plus the proper computing capacity could be programmed to accomplish these kinds of applications. The procedure would begin by deciding on the desired force to be applied in each axis direction. The controller would perform the following sequence of operations, with the resulting offset force calculated as illustrated in Fig. 6.2:

1. Measure the forces at the wrist in each axis direction.
2. Calculate the force offsets required. The force offset in each direction is determined by subtracting the desired force from the measured force.
3. Calculate the torques to be applied by each axis to generate the desired force offsets at the wrist. These are moment calculations which take into account the combined effects of the various joints and links of the robot.
4. Then the robot must provide the torques calculated in step 3 so that the desired forces are applied in each direction.

Force-sensing wrists are usually very rigid devices so that they will not deflect undesirably while under load. When designing a force-sensing wrist there are several problems that may be encountered. The end-of-the-arm is often in a relatively hostile environment. This means that the device must be sufficiently rugged to withstand the environment. For example, it must be capable of tolerating an occasional crash of the robot arm. At the same time the device must be sensitive enough to detect small forces. This design problem is usually solved by using overtravel limits. An overtravel limit is a physical stop designed to prevent the force sensor from deflecting so far that it would be damaged.

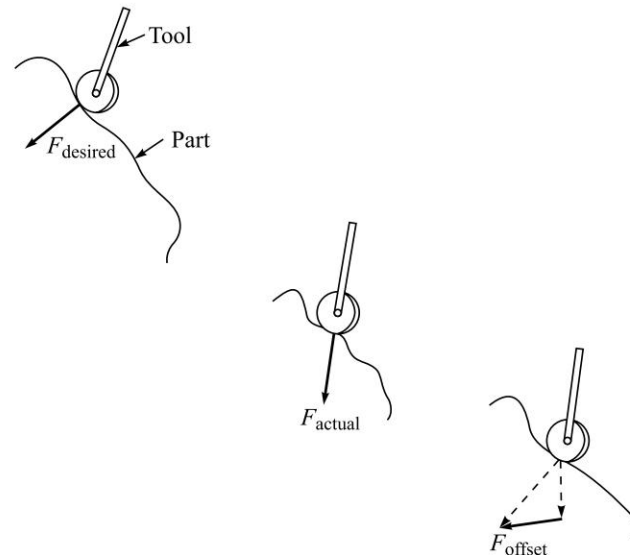


Fig. 6.2 Force accommodation, showing how the required offset force would compensate for the difference between actual force and desired force.

The calculations required to utilize a force-sensing wrist are complex and require considerable computation time. Also, for an arm traveling at moderate-to-high speeds, the level of control over the arm just as it makes contact with an object is limited by the dynamic performance of the arm. The momentum of the arm makes it difficult to stop its forward motion quickly enough to prevent a crash.

Design of force sensors is a very complex process due to the redundancies present in the force sensing process itself. Force sensors are made using strain gages that measure the strain along particular axes. As shown in the Fig. 6.3 a single cantilever beam can be used to measure forces along two axes directions, i.e., x -axis and z -axis.

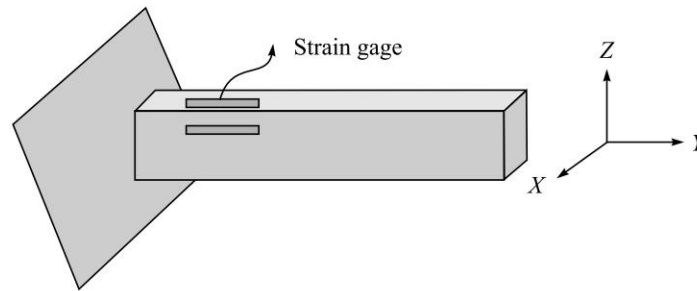


Fig. 6.3 Single cantilever beam fixed at one end and subjected to a load at the other end.

It is to be noted that in 3D space we can have a total of three forces and three moments. Hence, a single cantilever beam cannot measure all the six components. A force sensor is usually made having four arms, each of which responds to two forces as given in Fig. 6.4.

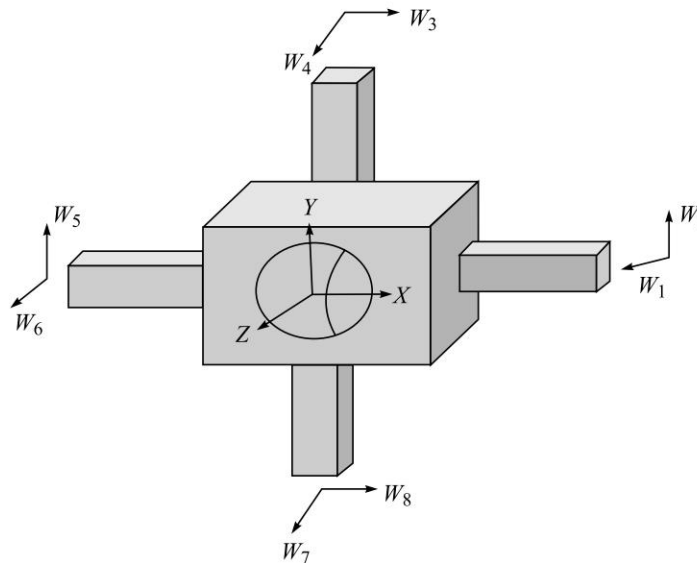


Fig. 6.4 Internal structure of sensor.

A complete force/torque sensor also called a wrist sensor is as shown in Fig. 6.5.

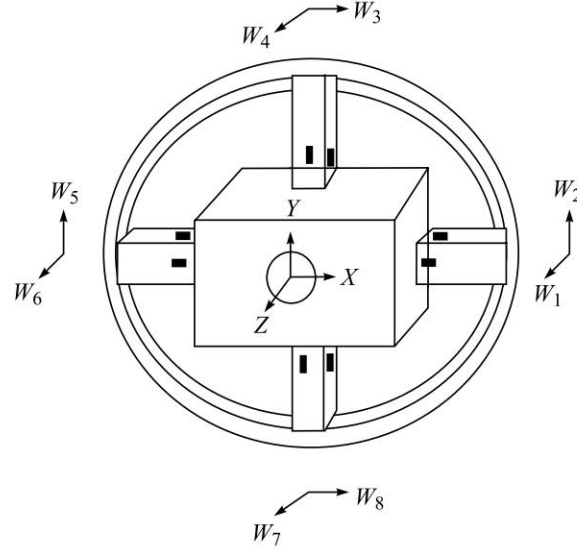


Fig. 6.5 A force torque sensor.

At the center, we have three forces and three moments acting. Writing the force balance equation for static equilibrium we have.

$$\begin{pmatrix} F_x \\ F_y \\ F_z \\ M_x \\ M_y \\ M_z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \\ w_7 \\ w_8 \end{pmatrix}$$

Writing these set of equation in short form, we have

$$F = CW$$

where matrix ' F ' contains the external forces and moments acting at the centre, ' C ' in the matrix relating the external forces to the forces at the beams and ' W ' is the strain gage readings corresponding to the forces in each beam. As can be seen from the equation above the matrix ' C ' is non-square and hence a unique

solution does not exist to obtain the value of $w_1 \dots w_8$, for a given force and moment input. Hence, force sensors need to be accurately calibrated before use and they also contain some errors as exact solution to the equation do not exist. The sensor is normally installed at the wrist of the robot. The forces and moments acting on the gripper is then reflected at the wrist sensor.

Joint sensing If the robot uses dc servomotors then the torque being exerted by the motors is proportional to the current flowing through the armature. A simple way to measure this current is to measure the voltage drop across a small precision resistor in series with the motor and power amplifier. This simplicity makes this technique attractive; however, measuring the joint torque has several disadvantages. First, measurements are made in joint space, while the forces of interest are applied by the tool and would be more useful if made in tool space. The measurements therefore not only reflect the forces being applied at the tool, but also the forces and torques required to accelerate the links of the arm and to overcome the friction and transmission losses of the joints. In fact, if the joint friction is relatively high (and it usually is), it will mask out the small forces being applied at the tool tip. One area where joint torque sensing shows promise of working well is with direct-drive robots. Direct-drive robots are a relatively new innovation in which the drive motors are located at the joints of the manipulator. In torque sensing, this configuration reduces the friction and transmission losses, and the problems of torque measurement which accompany these losses are thereby reduced.

Tactile array sensors A tactile array sensor is a special type of force sensor composed of a matrix of force-sensing elements. The force data provided by this type of device may be combined with pattern recognition techniques to describe a number of characteristics about the impression contacting the array sensor surface. Among these characteristics are 1. the presence of an object, 2. the object's contact area, shape, location, and orientation, 3. the pressure and pressure distribution, and 4. force magnitude and location. Tactile array sensors can be mounted in the fingers of the robot gripper or attached to a work table as a flat touch surface.

The device is typically composed of an array of conductive elastomer pads. As each pad is squeezed its electrical resistance changes in response to the amount of deflection in the pad, which is proportional to the applied force. By measuring the resistance of each pad, information about the shape of the object against the array of sensing elements can be determined. The operation of a tactile array sensor (with an 8×8 matrix of pressure-sensitive pads) is pictured in Fig. 6.6. In the background is the CRT monitor display of the tactile impression made by the object placed on the surface of the sensor device, As the number of pads in the array is increased the resolution of the displayed information improves.

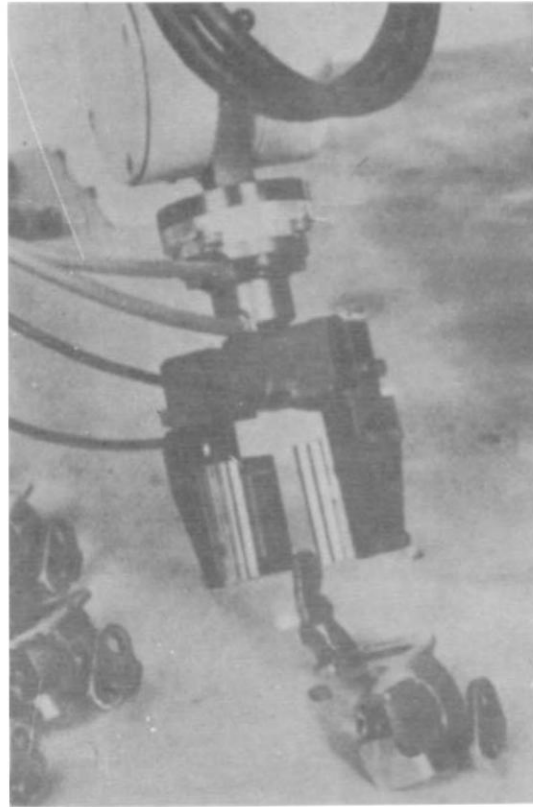


Fig. 6.6 Tactile array sensor device mounted in a mechanical gripper. (Photo courtesy: of Lord Corporation)

Example 6.1

A possible use of a tactile array sensor is to measure the force and moments being applied to an object by the robot. A typical application of this capability is the case of a robot inserting a peg into a hole. Suppose the peg misses the hole slightly creating a binding action between the peg and the hole. Figure 6.7 illustrates the situation showing the likely pattern of forces on the tactile array sensor surface.

The problems in designing and using a tactile array sensor include wear of the contacting surface and the possible need for multiplexing to minimize the number of interconnections if the number of sensors in the array is very large. Figure 6.8 shows the Lord Corporation's LTS 300 Series product which contains a large touch field surface (approximately 6 in. square) composed of an 80 by 80 pattern of sensor elements, making a total of 6400 sensitive sites. Each sensor has gray scale capability to allow force magnitude to be measured at each sensor location on the surface. The principal application foreseen for this product is for location and orientation of components in robotic assembly operations as shown in Fig. 6.9.



Fig. 6.7 Tactile array sensor mounted on a flat work surface. (Photo courtesy: Lord Corporation)

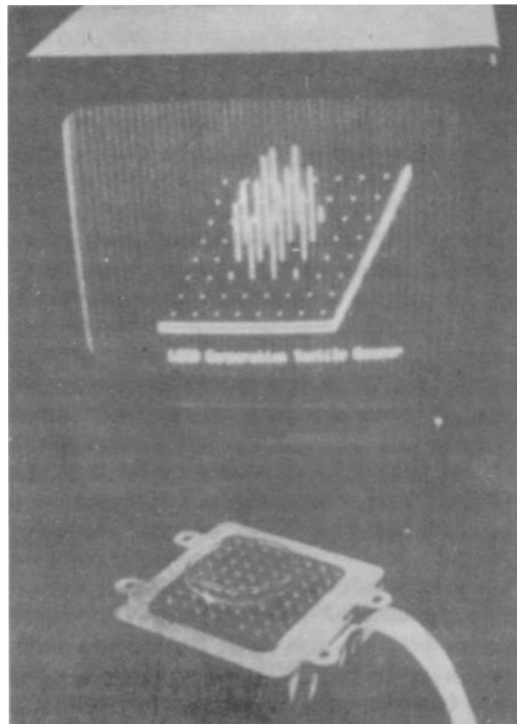


Fig. 6.8 Tactile array sensor using 8×8 sensor array with display of tactile impression. (Photo courtesy: Lord Corporation)

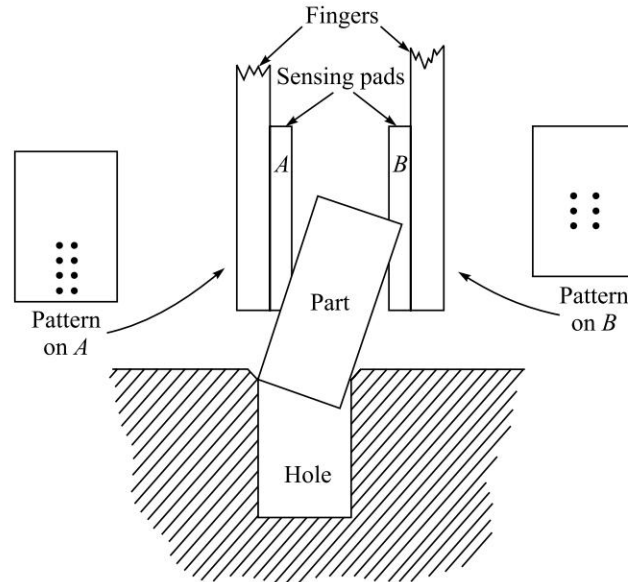


Fig. 6.9 Possible binding action in an insertion task.

Research into potential materials for tactile sensors has led to the development of a force-sensing skin of Polyvinylidene di-Fluoride. This is a piezoelectric material which means that it generates an output voltage when it is squeezed. Advantages of Polyvinylidene di-Fluoride include its availability in sheets only a few ten-thousandths of an inch and the fact that it is impervious to most chemicals. Another interesting feature of this material is that it is pyroelectric, meaning that it generates a voltage when heated. This feature suggests the possibility for designing a tactile sensor that provides simultaneous force and temperature sensing.

6.4 PROXIMITY AND RANGE SENSORS

Proximity sensors are devices that indicate when one object is close to another object. How close the object must be in order to activate the sensor is dependent on the particular device. The distances can be anywhere between several millimeters and several feet. Some of these sensors can also be used to measure the distance between the object and the sensor, and these devices are called range sensors. Proximity and range sensors would typically be located on the wrist or end effector since these are the moving parts of the robot. One practical use of a proximity sensor in robotics would be to detect the presence or absence of a workpart or other object. Another important application is for sensing human beings in the robot workcell. Range sensors would be useful for determining the location of an object (e.g., the workpart) in relation to the robot.

A variety of technologies are available for designing proximity and range sensors. These technologies include optical devices, acoustics, electrical field techniques (e.g., eddy currents and magnetic fields), and others. We will survey only a few of the possibilities in the following paragraphs.

Optical proximity sensors can be designed using either visible or invisible (infrared) light sources. Infrared sensors may be active or passive. The active sensors send out an infrared beam and respond to the reflection of the beam against a target. The infrared-reflectance sensor using an incandescent light source is a common device that is commercially available. The active infrared sensor can be used to indicate not only whether or not a part is present, but also the position of the part. By timing the interval from when the signal is sent and the echo is received, a measurement of the distance between the object and the sensor can be made. This feature is especially useful for locomotion and guidance systems. Passive infrared sensors are simply devices which detect the presence of infrared radiation in the environment. They are often utilized in security systems to detect the presence of bodies giving off heat within the range of the sensor. These sensor systems are effective at covering large areas in building interiors.

Another optical approach for proximity sensing involves the use of a collimated light beam and a linear array of light sensors. By reflecting the light beam off the surface of the object, the location of the object can be determined from the position of its reflected beam on the sensor array. This scheme is illustrated in Fig. 6.10. The formula for the distance between the object and the sensor is given as follows:

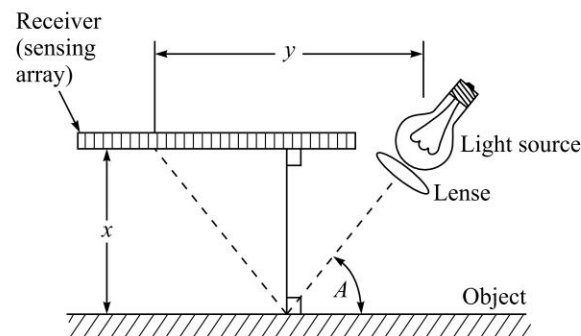


Fig. 6.10 Scheme for a proximity sensor using reflected light against a sensor array.

$$x = 0.5 y \tan(A)$$

where x = the distance of the object from the sensor

y = the lateral distance between the light source and the reflected light beam against the linear array. This distance corresponds to the number of elements contained within the reflected beam in the sensor array

A = the angle between the object and the sensor array as illustrated in Fig. 6.10.

Use of this device in the configuration shown relies on the fact that the surface of the object must be parallel to the sensing array.

Acoustical devices can be used as proximity sensors. Ultrasonic frequencies (above 20,000 Hz) are often used in these devices because the sound is beyond the

range of human hearing. One type of acoustical proximity sensor uses a cylindrical open-ended chamber with an acoustic emitter at the closed end of the chamber. The emitter sets up a pattern of standing waves in the cavity which is altered by the presence of an object near the open end. A microphone located in the wall of the chamber is used to sense the change in the sound pattern. This kind of device can also be used as a range sensor.

Proximity and range sensors based on the use of electrical fields are commercially available. Two of the types in this category are eddy-current sensors and magnetic field sensors. Eddy-current devices create a primary alternating magnetic field in the small region near the probe. This field induces eddy currents in an object placed in the region so long as the object is made of a conductive material. These eddy currents produce their own magnetic field which interacts with the primary field to change its flux density. The probe detects the change in the flux density and this indicates the presence of the object.

Magnetic field proximity sensors are relatively simple and can be made using a reed switch and a permanent magnet. The magnet can be made a part of the object being detected or it can be part of the sensor device. In either case, the device can be designed so that the presence of the object in the region of the sensor completes the magnetic circuit and activates the reed switch. This type of proximity sensor design is attractive because of its relative simplicity and because no external power supply is required for its operation.

6.5 MISCELLANEOUS SENSORS AND SENSOR BASED SYSTEMS

The miscellaneous category covers the remaining types of sensors and transducers that might be used for interlocks and other purposes in robotic workcells. This category includes devices with the capability to sense variables such as temperature, pressure, fluid flow, and electrical properties. Many of the common transducers and sensors used for these variables are listed in Table 6.2.

An area of robotics research that might be included in this chapter is voice sensing or voice programming. Voice-programming systems can be used in robotics for oral communication of instructions to the robot. Voice sensing relies on the techniques of speech recognition to analyze spoken words uttered by a human and compare those words with a set of stored word patterns. When the spoken word matches the stored word pattern, this indicates that the robot should perform some particular actions which correspond to the word or series of words. We will explore this technology in more detail in Chap. 19.

6.6 USES OF SENSORS IN ROBOTICS

The major uses of sensors in industrial robotics and other automated manufacturing systems can be divided into four basic categories:

1. Safety monitoring
2. Interlocks in workcell control

3. Part inspection for quality control
4. Determining positions and related information about objects in the robot cell

One of the important applications of sensor technology in automated manufacturing operations is safety or hazard monitoring which concerns the protection of human workers who work in the vicinity of the robot or other equipment. This subject of safety monitoring and other methods of ensuring worker safety in robotics is explored in Chap. 17.

The second major use of sensor technology in robotics is to implement interlocks in workcell control. As mentioned in Chap. 2, interlocks are used to coordinate the sequence of activities of the different pieces of equipment in the workcell. In the execution of the robot program, there are certain elements of the work cycle whose completion must be verified before proceeding with the next element in the cycle. Sensors, often very simple devices, are utilized to provide this kind of verification. We will discuss interlocks and their uses in Chap. 11 on workcell control.

The third category is quality control. Sensors can be used to determine a variety of part quality characteristics. Traditionally, quality control has been performed using manual inspection techniques on a statistical sampling basis. The use of sensors permits the inspection operation to be performed automatically on a 100 percent basis, in which every part is inspected. The limitation on the use of automatic inspection is that the sensor system can only inspect for a limited range of part characteristics and defects. For example, a sensor probe designed to measure part length cannot detect flaws in the part surface. Many applications of automated inspection are accomplished without the use of robotics. The reason for including this category in our discussion of robotic sensors is that robots are, in fact, often used to implement automatic inspection systems by means of sensors. We will discuss robot inspection applications in Chap. 15.

The fourth major use of sensors in robotics is to determine the positions and other information about various objects in the workcell (e.g., workparts, fixtures, people, equipment, etc.). In addition to positional data about a particular object, other information required to properly execute the work cycle might include the object's orientation, color, size, and other characteristics. Reasons why this kind of data would need to be determined during the program execution include:

- Workpart identification, and
- Random position and orientation of parts in the workcell

Accuracy requirements in a given application exceed the inherent capabilities of the robot. Feedback information is required to improve the accuracy of the robot's positioning.

An example of workpart identification would be in a workcell in which the robot processes several types of workparts, each requiring a different sequence of actions by the robot. Each part presented to the robot would have to be properly identified so that the correct subroutine could be called for execution. This type of identification problem arises in automobile body spot-welding lines where the line is designed to weld several different body styles (e.g., coupes, sedans, wagons). Each welding robot along the line must execute the welding cycle for the particular body style at that station. Simple optical sensors are typically used to indicate the presence or absence of specific body style features in order to make the proper identification.

An example of the part position and orientation problem is where a robot would be required to pick up parts moving along a conveyor in random orientation and position, and place them into a fixture. To accomplish the task, the exact location of each part would have to be sensed as it came down the line. In addition, for the robot to use a mechanical gripper to grasp the particular workpart, the orientation of the part on the conveyor would have to be determined. All of this information would have to be processed by the workcell controller (or other computer) in real time in order to guide the robot in the execution of its programmed work cycle. Vision systems represent an important category of sensor system that might be employed to determine such characteristics as part location and orientation. We consider machine vision to be a topic in robotics which is significant enough to merit its own chapter. That chapter follows this one.

In some applications, the accuracy requirements in the application are more stringent than the inherent accuracy and repeatability of the robot. Certain assembly operations represent examples of this case. The robot is required to assemble two parts whose alignment must be very close, closer than the accuracy of the robot. One possible solution that might be used is a remote center compliance (RCC) device. The use of the RCC device in assembly will be described in Chap. 15.

All four categories of sensor applications (safety monitoring, interlocks, inspection, and positional data) are instances where the sensor constitutes a component of a control system used in the robot work cell to accomplish some specific control function. That control system, in turn, is a component of a larger control system which we are calling the workcell control system. All of the control functions which takes place in the workcell are coordinated and regulated by this larger system. Our discussion of robot workcell control will resume in Chap. 11, after we have examined robot programming, an obvious prerequisite for workcell control.

Problems

- 6.1** It is desired to design a safety monitoring system for a robot cell in which a robot loads an automatic production machine with parts arriving on a conveyor. The robot is large and it is considered dangerous for workers to wander into the work volume of the robot. Aside from the other safety precautions that might be taken to ensure the safety of the workers, the safety monitoring system will use one or more sensors to detect the presence of humans in the cell.
- (a)** Write the detailed 'functional specifications' for the sensor system. That is, make up a list of the things the sensor system must do, and the ways in which it will have to operate.
 - (b)** From the list of sensors in Table 6.2, select several alternative sensors that will satisfy the functional specifications, and compare its features against the specifications. For each sensor, explain how it will be configured in the safety monitoring system. Use sketches if necessary to illustrate the configuration.
 - (c)** Select the best sensor alternative, and justify your selection.

CHAPTER 7

Actuators and Drive Systems

7.1 Introduction

Actuators are the muscles of robots. If the links and the joints are the skeleton of the robot, the actuators act as muscles that move or rotate the links to change the configuration of robots. The actuator must have enough power to accelerate and decelerate the links and to carry the loads and yet be light, economical, accurate, responsive, reliable, and easy to maintain.

There are many types of actuators available and, undoubtedly, there will be more varieties available in the future. At least, the following types are worth mentioning:

- Electric motors
 - Servomotors
 - Stepper motors
 - Direct drive electric motors
- Hydraulic actuators
- Pneumatic actuators
- Novelty actuators

Electric motors, and specially servomotors, are the most commonly used robotic actuators. Hydraulic systems were very popular for large robots in the past and are still around in many places but are no longer as popular, except for large applications. Pneumatic actuators are used in robots that have 1/2 degree of freedom, on-off type joints, as well as for insertion purposes. Novelty actuators, including direct drive electric motors, electroactive polymer actuators, muscle-wire actuators, and piezoelectric actuators, are mostly used in research and development work or as specialty items for specific purposes but may become more useful in the near future.

In the following section, we will compare the common characteristics of different types of actuators, followed by the individual study of each type.

7.2 Characteristics of Actuating Systems

The following characteristics may be used to compare different actuating systems. In addition to these, depending on the special circumstances in which they will be used, other characteristics may play a role in the design of robots. Examples include underwater systems, where waterproof operation of a system is very important, and space systems, where the lift-off weight and reliability are of absolute importance.

7.2.1 Nominal Characteristics—Weight, Power to Weight Ratio, Operating Pressure, Voltage, and Others

It is important to consider the nominal characteristics of actuators. These include weight, power and power to weight ratio, operating pressure, operating voltage, temperatures, and others. For example, since in many robotic systems the actuators are placed directly at the joints, and therefore, move with them, the weight of the actuator acts as a load on the preceding actuators and must be accelerated and decelerated by them. A heavier actuator downstream requires more torque upstream, resulting in larger power requirements and heavier actuators. Consequently, it is very important to consider the weight and placement of actuators.

Another important characteristic is power to weight ratio. For example, the power to weight ratio of electric systems is average. Stepper motors are generally heavier than servomotors for the same power, and therefore, have a lower power to weight ratio. Hydraulic systems have the highest power to weight ratio. However, it is important to realize that in these systems, the weight is actually composed of two portions. One is the hydraulic actuator, and the other is the hydraulic power unit. The system's power unit consists of a pump that generates the high pressure needed to operate the actuator, a reservoir, filters, electric drive motors to drive the pump, cooling units, valves, and so on. However, the power unit is normally stationary, somewhere away from the robot itself. It does not move with the actuator. The power is brought to the robot via an umbilical tether hose. Consequently, the actual power to weight ratio of the actuator is very high for the moving parts. If the power unit must also move with the robot (e.g., a hydraulic transportation robot), the total power to weight ratio will be much less. Pneumatic actuators deliver the lowest power to weight ratio. The power the hydraulic system delivers is very high due to high operating pressures. This may range from 50 psi to 5000 psi. Pneumatic cylinders normally operate around 100 to 120 psi. The higher pressures in hydraulic systems create higher power concentrations, but also require higher maintenance; if a leak occurs, they can be more dangerous.

Electric motors that operate at a higher voltage have a better power to weight ratio, too. Additionally, as we will see later, for the same power output, as the voltage to an electric motor increases the required current will decrease, reducing the size of required wires. The heat generated in the motor is a second-order function of the current; therefore, as the current decreases, the generated heat decreases and efficiency is increased.

7.2.2 Stiffness versus Compliance

Stiffness is the resistance of a material against deformation. It may be the stiffness of a beam against bending under the load, the resistance of a gas against compression in a

cylinder under load, or resistance of wine against compression in a bottle during corking operation. A stiffer system requires a larger load to deform. Conversely, a more compliant system requires a smaller load to deform.

Stiffness is directly related to the modulus of elasticity of the material. The modulus of elasticity of fluids can be around 1×10^6 psi, which is very high. As a result, hydraulic systems are very stiff and noncompliant. Conversely, because air is compressible, pneumatic systems are compliant.

Stiff systems have a more rapid response to changing loads and pressures and are more accurate. Obviously, if a system is compliant, it can easily deform (or compress) under changing load or changing driving force; consequently, it is inaccurate. Similarly, if a small driving force is applied to a hydraulic ram, due to its stiffness, it responds more rapidly and more accurately than a pneumatic system that deforms under the load. Additionally, a stiff system resists deformation under the load, and therefore, holds its position more accurately. Now consider a robot that is used to insert an IC chip into a circuit board. If the system is not stiff enough, it will not be able to push the chip into the board because the actuator may deform under the resistive force. On the other hand, if the part and the holes are not perfectly aligned, a stiff system does not deform enough to prevent damage to the robot or the part, whereas a compliant system gives to prevent damage. So, although stiffness causes a more responsive and more accurate system, it also creates a danger if all things are not always perfect. Consequently, a working balance is needed between these two competing characteristics. We will discuss a particular solution to this problem called Remote-Center Compliance (RCC) device later.

7.2.3 Use of Reduction Gears

Some systems, such as hydraulic actuators and direct-drive electric motors, produce very large forces or torques with short strokes. This means that the actuator may be moved very slightly while delivering its full force or torque. As a result, there is no need to use reduction gear trains to increase the torque it produces and to slow it down to manageable speeds. For this reason, hydraulic actuators can be directly attached to the links, which simplify the design, reduce the weight, cost, and rotating inertia of joints, reduce backlash, increase reliability of the system due to simpler design and fewer parts, and reduce noise. On the other hand, electric motors rotate at high speeds, up to many thousands of revolutions per minute, and must be used in conjunction with reduction gears to increase their torque and decrease their speed, as no one would want a robot arm to be rotating at such high speeds. This, of course, increases the cost, number of parts, backlash, inertia of the rotating body, and so on, as we discussed earlier, but also increases the resolution of the system, as it is possible to rotate the link a very small angle.

Now suppose that, through a set of reduction gears with a reduction ratio of N , a load with inertia I_l is connected to a motor with inertia I_m (including the inertia of the reduction gears), as shown in Figure 7.1. The torque and speed ratio between the motor and the load are:

$$\begin{aligned} T_l &= NT_m \\ \dot{\theta}_l &= \frac{1}{N} \dot{\theta}_m \quad \text{and} \quad \ddot{\theta}_l = \frac{1}{N} \ddot{\theta}_m \end{aligned} \quad (7.1)$$

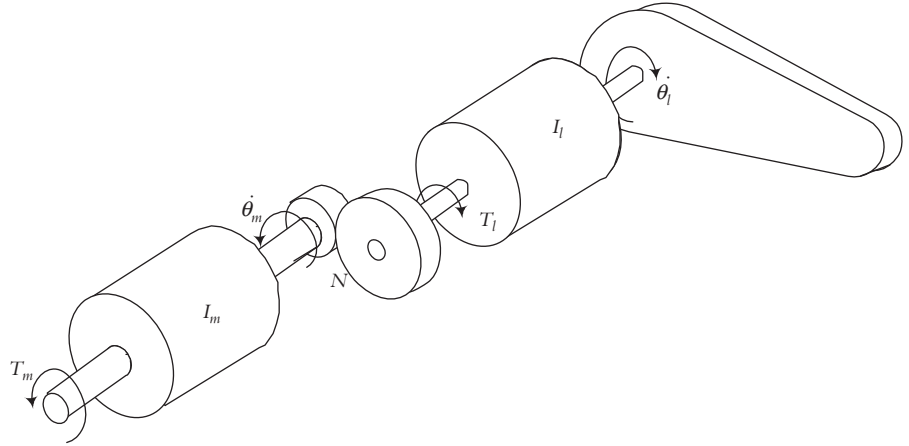


Figure 7.1 Inertia and torque relationship between a motor and a load.

If we write the torque balance equation for the system from free-body-diagrams of Figure 7.2, and substitute from Equation (7.1), we get:

$$\begin{aligned}
 T_m - \frac{1}{N} T_l &= I_m \ddot{\theta}_m + b_m \dot{\theta}_m \quad \text{and} \quad T_l = I_l \ddot{\theta}_l + b_l \dot{\theta}_l \\
 T_m &= I_m \ddot{\theta}_m + b_m \dot{\theta}_m + \frac{1}{N} (I_l \ddot{\theta}_l + b_l \dot{\theta}_l) \\
 T_m &= I_m \ddot{\theta}_m + b_m \dot{\theta}_m + \frac{1}{N^2} (I_l \ddot{\theta}_m + b_l \dot{\theta}_m)
 \end{aligned} \tag{7.2}$$

where b_m and b_l are viscous coefficients of friction for the motor and the load.

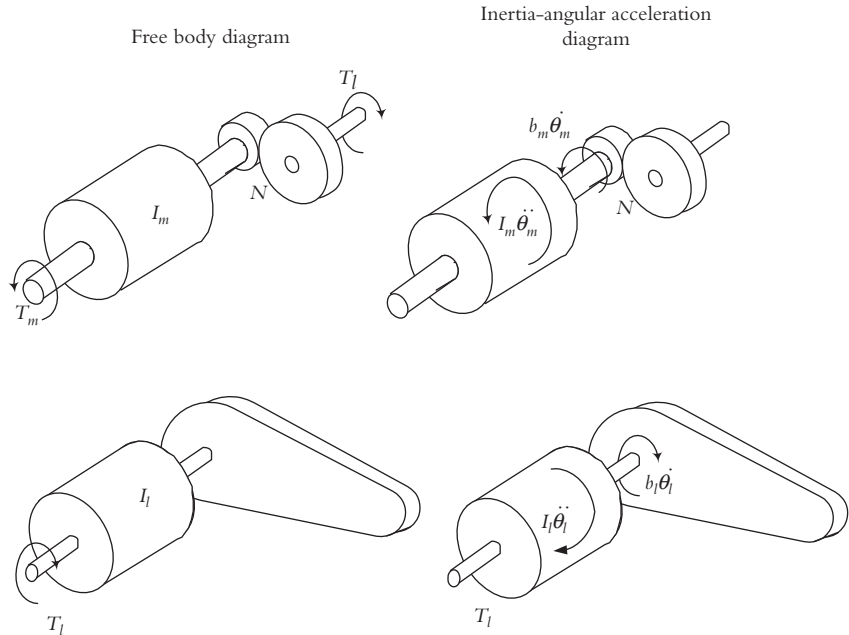


Figure 7.2 Free body diagrams of the motor and the load.

As Equation (7.2) indicates, the effective inertia of the load felt by the motor is conversely proportional to the square of the reduction gear ratio, or:

$$I_{Effective} = \frac{1}{N^2} I_l \quad \text{and} \quad I_{Total} = \frac{1}{N^2} I_l + I_m \quad (7.3)$$

So, the motor will only “feel” a fraction of the actual inertia of the load, which, in the case of a robot, constitutes both the manipulator and the load it carries. Reduction ratios of 20–100 are common in manipulator robots. Therefore, the total inertia applied to the motor from the load may only be 1/400th to 1/10,000th of the actual inertia, allowing the motor to accelerate and decelerate quickly. In direct drive systems, both electric and hydraulic, the actuators are exposed to the full inertial loads. With high gear ratios, the inertial effects of the load can actually be ignored in the control system of the robot.

Please note that the opposite is also true, that the effect of the inertia of the motor on the load is also 400–10,000 times larger. To reduce this effect, designers opt to use low-inertia motors with long and slender rotors or pancake motors.

Example 7.1

A motor with rotor inertia of 0.015 Kgm² and maximum torque of 8 Nm is connected to a uniformly distributed arm with a concentrated mass at its end, as shown in Figure 7.3. Ignoring the inertia of a pair of reduction gears and viscous friction in the system, calculate the total inertia felt by the motor and the maximum angular acceleration it can develop if the gear ratio is (a) 3 or (b) 30.

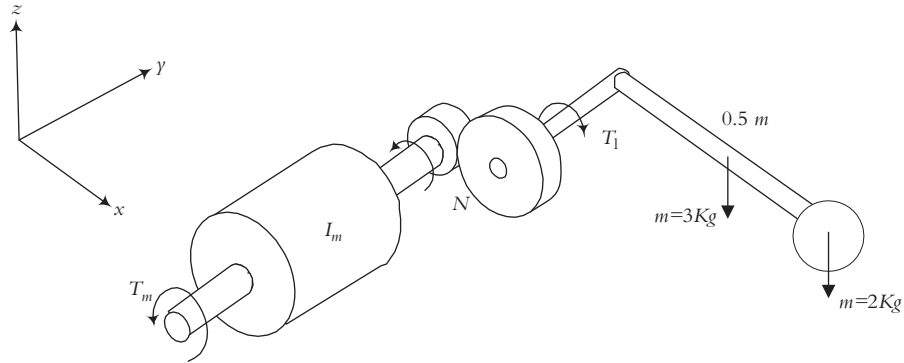


Figure 7.3 Schematic drawing of the system of Example 7.1.

Solution: This is very similar to a robot arm and a servomotor actuator. The total moment of inertia of the arm and the concentrated mass at the center of rotation is:

$$\begin{aligned} I_l &= I_{arm} + I_{mass} = \frac{1}{3} m_{arm} l^2 + m_{mass} l^2 \\ &= \frac{1}{3} (3) (0.5)^2 + (2) (0.5)^2 = 0.75 \text{ Kgm}^2 \end{aligned}$$

From Equation (7.3):

$$(a) \quad I_{Total} = \frac{1}{N^2} I_l + I_m = \frac{1}{9} (0.75) + 0.015 = 0.098 \text{ Kgm}^2$$

$$(b) \quad I_{Total} = \frac{1}{900} (0.75) + 0.015 = 0.0158 \text{ Kgm}^2$$

As you can see, the total inertia with the higher gear reduction ratio is practically the same as the rotor inertia of the motor. The maximum angular accelerations will be:

$$(a) \quad \ddot{\theta}_m = \frac{T_m}{I_{total}} = \frac{8}{0.098} = 82 \text{ rad/sec}^2$$

$$(b) \quad \ddot{\theta}_m = \frac{T_m}{I_{total}} = \frac{8}{0.0158} = 506 \text{ rad/sec}^2$$

The no-load maximum angular acceleration of the motor would be about 530 rad/sec². ■

7.3 Comparison of Actuating Systems

Table 7.1 is a summary of actuator characteristics. We will refer to, and discuss, these characteristics throughout this chapter.

Table 7.1 *Summary of Actuator Characteristics.*

Hydraulic	Electric	Pneumatic
+ Good for large robots and heavy payload + Highest power/weight ratio + Stiff system, high accuracy, better response + No reduction gear needed + Can work in wide range of speeds without difficulty + Can be left in position without any damage – May leak; not fit for clean room applications – Requires pump, reservoir, motor, hoses, and so on – Can be expensive and noisy; requires maintenance – Viscosity of oil changes with temperature – Very susceptible to dirt and other foreign material in oil – Low compliance – High torque, high pressure, large inertia on the actuator	+ Good for all sizes of robots + Better control, good for high precision robots + Higher compliance than hydraulics + Reduction gears reduce inertia on the motor + Does not leak, good for clean room + Reliable, low maintenance + Can be spark-free; good for explosive environments – Low stiffness – Needs reduction gears, increased backlash, cost, weight, and so on – Motor needs braking device when not powered; otherwise, the arm will fall	+ Many components are usually off-the-shelf + Reliable components + No leaks or sparks + Inexpensive and simple + Low pressure compared to hydraulics + Good for on-off applications and for pick and place + Compliant systems – Noisy – Require pressurized air, filter, and so on – Difficult to control their linear position – Deform under load constantly – Very low stiffness; inaccurate response – Lowest power to weight ratio

7.4 Hydraulic Actuators

Hydraulic systems and actuators offer high power to weight ratio, large forces at low speeds, both linear and rotary actuation, compatibility with microprocessor and electronic controls, and tolerance of extreme hazardous environments. They can hold a load without need for a brake, generate less heat at the actuator, and apply a torque without the need for gearing. Many large robots of the past decades, mostly used in automobile production, were Cincinnati MilacronTM T3 hydraulic robots or other brands with similar characteristics. The T3 robot offered a payload of over 220 lb at 7 feet high, an impressive value. However, due to the leakage problem that is almost inevitable in hydraulic systems, and due to their power unit weight and cost, they are no longer common. Today, most robots are electric; however, still many robots in the industry have hydraulic actuators. Additionally, for special applications such as very large robots in civil and military service, hydraulic actuators may be the appropriate choice.

One important difference between hydraulic actuators and electric motors is that a hydraulic pump (not the actuator) may be sized for average load, whereas electric actuators must be sized for maximum load. This is because a hydraulic system uses an accumulator that stores the constant energy of the pump and can deliver larger loads when necessary. Therefore, if there are breaks in the motion, the accumulator stores energy needed for maximum loads. Another important consideration is that in general, an electric actuator must be located at or near a joint, adding to the mass and inertia of the robot. However, in hydraulic systems, only the actuator, the control valves, and the accumulators are near the joints. The hydraulic power unit may be located remotely, reducing the mass and inertia, especially when many joints are present.

The total force that a linear cylinder can deliver can be tremendously large for its size. A hydraulic cylinder can deliver a force of $F = p \times A$ lb, where A is the effective area of the piston or ram, and p is the working pressure. For example, for a 1000 psi pressure, every square inch of the cylinder develops 1000 lb of force. In rotary cylinders, the same principle is true, except that the output is a torque where:

$$dA = t \cdot dr$$

$$T = \int_{r_1}^{r_2} p \cdot r \cdot dA = \int_{r_1}^{r_2} p \cdot r \cdot t \cdot dr = pt \int_{r_1}^{r_2} r \cdot dr = \frac{1}{2}pt(r_2^2 - r_1^2) \quad (7.4)$$

where p is the fluid pressure, t is the thickness or width of the rotary cylinder, and r_1 and r_2 are the inner and outer radii of the rotary cylinder, as shown in Figure 7.4.

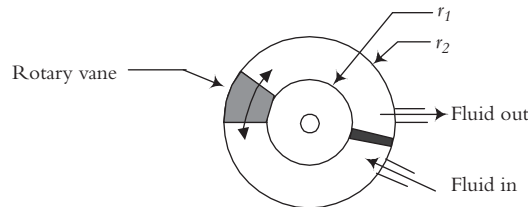


Figure 7.4 A rotary hydraulic actuator. This actuator can be directly attached to a revolute joint without any need for gear reduction.

The flow rate and volume of fluid needed in a hydraulic system are:

$$d(Vol) = \frac{\pi d^2}{4} dx \quad (7.5)$$

$$Q = \frac{d(Vol)}{dt} = \frac{\pi d^2}{4} \frac{dx}{dt} = \frac{\pi d^2}{4} \dot{x} \quad (7.6)$$

where dx is the desired displacement, and $\dot{x}$ is the desired velocity of the piston. By controlling the volume of the fluid going into the cylinder, the total displacement can be controlled. By controlling the rate at which the fluid is delivered to the cylinder, the velocity can be controlled. This is done through a servovalve that controls both the volume of the fluid as well its rate. We will discuss the servovalves later.

A hydraulic system generally consists of the following components or subsystems:

- Hydraulic linear or rotary actuators that provide the force or torque needed to move the joints and are controlled by the servovalves or manual valves.
- Hydraulic pump that provides high-pressure fluid to the system.
- Electric motor (or in cases such as in a mobile unit, an engine) that operates the hydraulic pump.
- Cooling system, which rids the system of the generated heat. In some systems, in addition to cooling fans, radiators and cooled air are used.
- Reservoir, which keeps the fluid supply available to the system. Since the pump is usually constantly supplying the system, whether the system is using it or not, the extra pressurized fluid and the returned fluid from the cylinders flow back into the reservoir.
- Accumulators that are used to store some extra energy for maximum loads, especially when the pump is designed for average use.
- Servovalves that control the amount and the rate of fluid to the cylinders. The servovalve is generally driven by a hydraulic servomotor.
- Check valves that are used for safety and controlling maximum pressure.
- Holding valves that are used to prevent the actuator from moving when the system is turned off or power is lost. Holding valves fulfill the need for a brake to prevent accidental motions due to power loss or when the system is shut down.
- Connecting hoses that are used to transport the pressurized fluid to the cylinders and back to the reservoir.
- Filtering system that maintains the quality and purity of the fluid. Due to its nature, moisture in the fluid damages hydraulic actuators and must be separated from the fluid.
- Sensors that are used as feedback to control the motion of the actuators. They include position, velocity, magnetic, touch, and other sensors. Additionally, other sensors are used for safety. For example, in case a servovalve is missing, the pump should not turn on. Otherwise, dangerous high-pressure fluid will squirt out of the ports.

Figure 7.5 is a schematic drawing of a typical hydraulic system. Refer to Chapter 6 for more about hydraulic controllers.

Figure 7.6 is a schematic drawing of a position control pilot valve for a hydraulic cylinder (also called a spool valve). This is a balanced valve, which means that the pressures on the two sides of the spool are equal. Therefore, it takes very little force to move the spool (due to friction), even though it may be under very high pressures. A

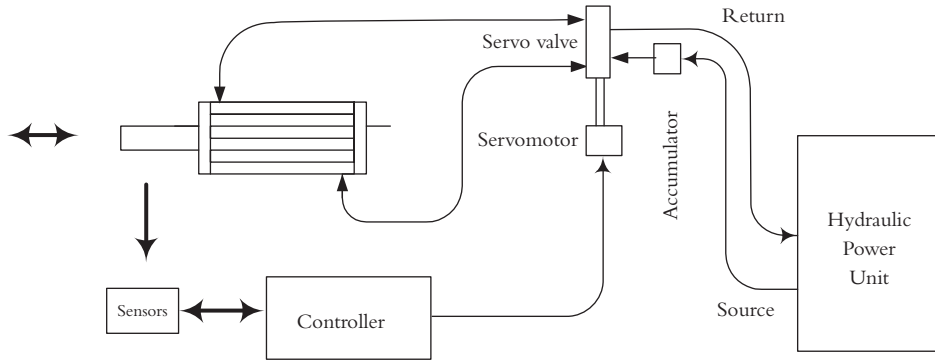


Figure 7.5 Schematic of a hydraulic system and its components.

spool valve may be operated manually or by a servomotor, called a servovalve. The servovalve and the cylinder together form a hydraulic actuator. As the spool moves up or down, it opens the supply and return ports through which, the fluid travels to the cylinder or is returned to the reservoir. Depending on the size of the opening of the port as the spool moves, the supply fluid flow rate is controlled, and so is the velocity of the cylinder. Depending on the length of time that the port is kept open, the total amount of the fluid to the cylinder, and consequently its total travel, is controlled. This can be written as:

$$q = Cx \quad (7.7)$$

$$\text{and } q(dt) = d(vol) = A(dy) \quad (7.8)$$

where q is the flow rate, C is a constant, x is the spool's displacement, A is the area of the piston, and y is piston's displacement. Combining Equations (7.7) and (7.8), and

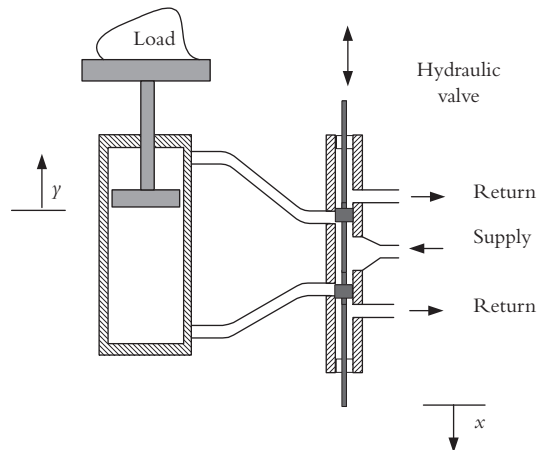


Figure 7.6 Schematic of a spool valve in neutral position.

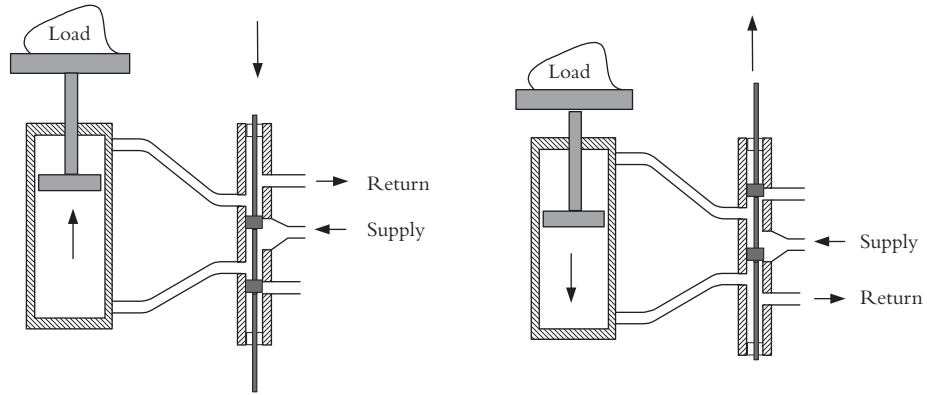


Figure 7.7 Schematic of a spool valve in open position. Depending on which ports are opened, the direction of motion of the piston will change.

designating d/dt as D , we get:

$$Cx(dt) = A(dy)$$

and

$$y = \frac{C}{AD}x \quad (7.9)$$

which shows that the hydraulic servomotor is an integrator. (Refer to Chapter 6 for more detail.)

The command to the servomotor controlling the spool valve comes from the controller. The controller sets the current to the servomotor as well as the duration the current is applied, which in turn, controls the position of the spool. Therefore, for a robot, when the controller has calculated how much and how fast a joint must move, it sets the current and its duration to the servomotor, which in turn, controls the position and rate of movement of the spool valve, controlling the flow of the fluid and its rate to the cylinder and moving the joint. The sensors provide feedback to the controller for accurate and continued control. Figure 7.7 shows the flow of the fluid as the spool valve moves up or down. As you can see, a simple motion of the spool controls the motion of the cylinder.

To provide feedback to the servovalve (otherwise it will not be a servovalve, but a manual spool valve), either electronic or mechanical feedback can be added (also refer to Figure 6.30). Figure 7.8 shows a simple mechanical feedback loop. A similar design is used in a two-stage spool valve to provide feedback. As you can see in Figure 7.8, a simple lever is added between the output and the input to provide an error signal to the system. As the desired position for the load is set by the setpoint lever (moved up, which will move down the load), the spool valve is opened, which will operate the cylinder. However, in reality, this lever arm is providing an error signal to the system, and therefore, the system responds by moving the cylinder. The error signal is integrated by the integrator (in this case, the cylinder), and as the error approaches zero, the signal to the system goes to zero. As the cylinder starts to move in the direction of the desired position (in this case, downward), the error signal becomes smaller which in turn reduces the output signal by closing the spool valve. By the time the load is at the desired position, the spool valve is closed and the load stops. Figure 7.9 is the schematic of the block diagram

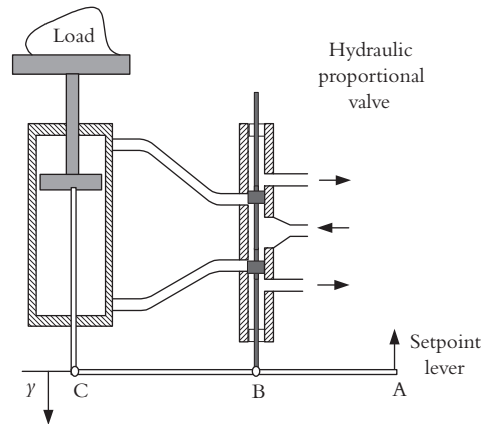


Figure 7.8 Schematic of a simple control device with proportional feedback.

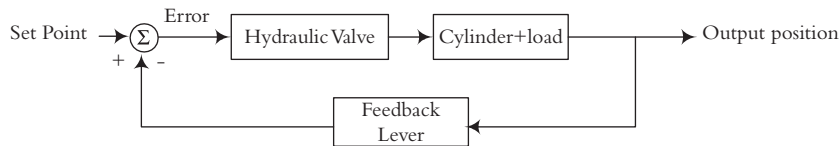


Figure 7.9 Block diagram of the hydraulic system with proportional feedback control scheme.

for this feedback loop. In reality, the feedback mechanism shown here is integrated into the servovalve.

There are many intricate and small passageways inside a hydraulic valve, especially in servovalves with feedback control integrated into the valve. In fact, this is why these systems are so susceptible to dirt or viscosity changes due to temperature. The smallest of foreign particles can affect the servovalve by restricting its passageways or ports. Similarly, viscosity changes can change the response of the valve as the fluid becomes less or more viscous. To visualize how these valves are constructed, imagine that you slice the valve into thin layers. Each layer will have corresponding portions of the passageways and ports and openings in it. We may manufacture the slices, which is easy to do by different methods such as a press, assemble the layers, and then either diffuse the layers together into one piece (under proper pressure and temperature), or connect them into one piece by through-bolts. This method is extensively used in industry to manufacture valves and many other products such as camera bodies. Figure 7.10 shows a simple example of this technique. When the thin slices are put together, a three-dimensional object results. Many rapid prototyping machines use the same technique to create 3-D prototypes of products.

Other designs have also been used for hydraulic actuation. For example, IBM 7565 robotic manufacturing system consisted of a gantry, 6-axis, hydraulic robot that was actuated at each of its three linear joints by a linear hydraulic motor. Each linear motor consisted of a set of four small hydraulic cylinders, which sequentially moved against a

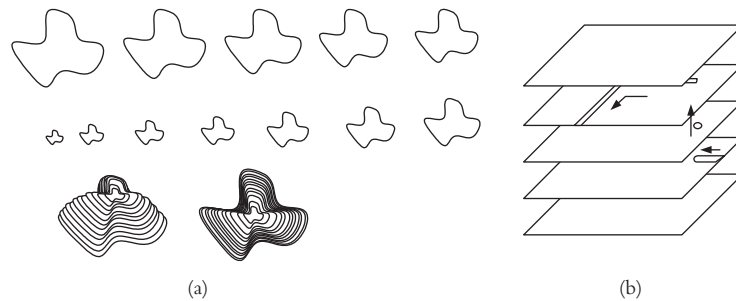


Figure 7.10 An object can be sliced into thin layers. When the layers are cut out individually and put together later, they will recreate the object. This method is used for rapid prototyping and for manufacturing complicated parts with intricate internal openings and passageways. (b) In this simple case, the fluid enters in the right, travels up one layer, and exits on lower left.

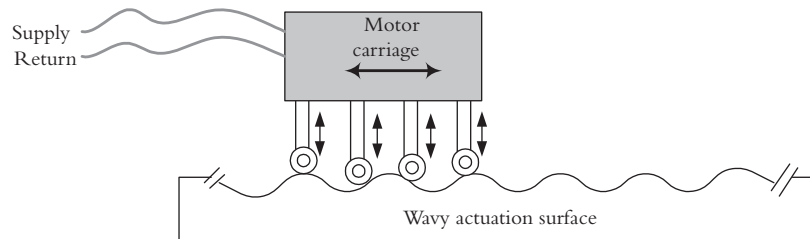


Figure 7.11 IBM 7565 linear hydraulic motor. As the four cylinders moved against the wavy actuation surface, the motor carriage moved sideways. (Drawn by S. Niku.)

wavy actuation surface, as shown in Figure 7.11. The four cylinders were forced into a position that corresponded to the desired position against the actuation surface. This in turn forced the motor carriage to move sideways. The four cylinders were controlled by a single servovalve. The advantage of this system was that by adding simple sections of the waved surface, the actuation could be made as long as desired. Figure 7.12 shows this actuator on the IBM 7565 robot.

Another hydraulic actuator is modeled after biological muscles. A muscle actuates a bone by contracting and, consequently, becoming shorter. In the hydraulic version, a similar design is used, where an oval shaped bladder is placed inside a shear sheath, as shown in Figure 7.13. When the pressure in the bladder increases, it becomes more spherical, causing the shear sheath to bulge out and shorten, just like a muscle. This design is promising, but due to nonlinearities inherent in the system and technical difficulties, it has not become practical yet. However, since it looks like a biological muscle, it can be very useful in humanoid robots.

Refer to our previous discussion in Chapter 6 regarding PID or other schemes for controlling a servovalve. All applicable theory and practice discussed there applies to servovalves, too.

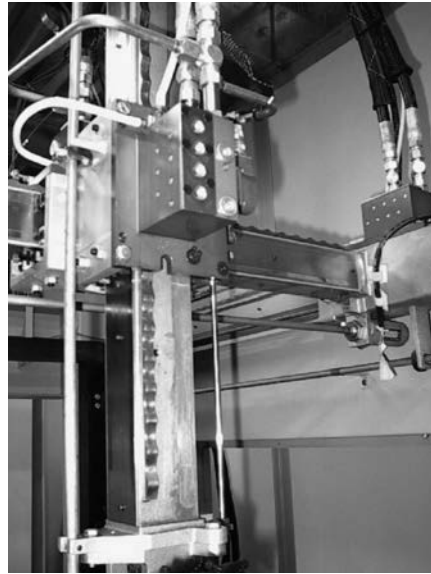


Figure 7.12 The linear hydraulic motor of IBM 7565 robot.

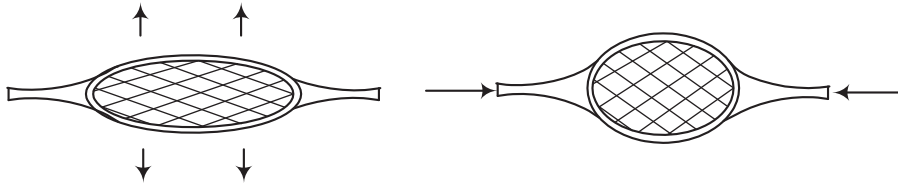


Figure 7.13 Schematic of a muscle type hydraulic actuator.

7.5 Pneumatic Devices

Pneumatic devices are principally very similar to hydraulic systems. A source of pressurized air is used to power and drive linear or rotary cylinders, controlled by manual or electrically controlled solenoid valves. Since the source of pressurized air is separate from the moving actuators, these systems have lower inertial loads. However, since pneumatic devices operate at a much lower air pressure, usually up to 100–120 psi max, their power to weight ratio is much lower than hydraulic systems.

The major problem with pneumatic devices is that air is compressible and, as a result, its volume changes under load. Consequently, pneumatic actuators are usually only used for insertion purposes, where the actuator is all the way forward or all the way backward, or they are used with 1/2-DOF joints that are fully on or fully off. Otherwise, controlling the exact position of pneumatic cylinders is very difficult.

One way to control the displacement of the pneumatic cylinders is called differential dithering. In this system, the exact location of the piston is sensed by a feedback sensor such as a linear encoder or potentiometer. This information is used in a controller that controls the air pressure on the two sides of the cylinder through a servovalve to control the exact position.¹

Pneumatic actuators are simple, rugged, and safe. Even if they leak, the air is not a contaminant. Most components are off the shelf, and therefore, easy to use and inexpensive. They are mostly used either as on/off devices or as accessories in a robotic cell in conjunction with robots for material handling and similar purposes.

7.6 Electric Motors

When a wire carrying a current is placed within a magnetic field, it will experience a force normal to the plane formed by the magnetic field and the current. Therefore, each side of the wire (coil) experiences a force as shown in Figure 7.14. If the wire is attached to a center of rotation, the resulting torque will cause it to rotate about the center of rotation. Changing the direction of the magnetic field or the current will cause a change in the direction of the force and the wire will continue to rotate about the center of rotation. As long as this change continues, the coil will continue to rotate. In practice, in order to accomplish this change in the current, either a set of commutators and brushes or slip rings are used for DC motors, the current is electronically switched for DC brushless motors, or AC current is used for AC motors, although in this case the permanent magnets and the coil are switched.² This is the basic principle behind all electric motors. Similarly, if a conductor is moved within a magnetic field crossing the flux, a current is induced in the conductor like a generator. This is called back-electromotive-force, or back-emf.

There are many types of motors, including AC induction motors, AC synchronous motors, DC brushed motors, DC brushless motors, stepper motors, direct drive DC motors, switched reluctance motors, AC/DC universal motors, and other varieties such as three-phase AC motors, disk motors, and others. Although we will discuss a number of issues about electric motors, the assumption is that you have already studied about different motors and how they operate. So, the discussion here will be at a minimum and only about the subjects directly related to robotic actuation. Please refer to other sources for additional detail about motors and their drive circuitry.

Except for stepper motors, all other types of motors can be used as a servomotor, which will be discussed later. In each case, the torque or power output of the motor is a function of the strength of the magnetic fields and the current in the windings as well as the length of the conductors of the coils. Some motors have permanent magnets (PM). These motors generate less heat, since the field is always present and no current is needed to build it.

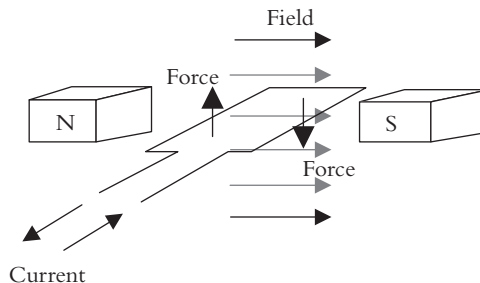


Figure 7.14 A wire carrying a current, placed within a magnetic field, will experience a force in a direction normal to a plane formed by the current and the field.

Others have a soft iron core and windings, where an electric current creates the magnetic field. In this case, more heat is generated, but when needed, the magnetic field can be varied by changing the current, whereas in permanent magnet motors, the field is constant. Additionally, under certain conditions, it is possible that the permanent magnet may get damaged and lose its field strength, in which case, the motor becomes useless.

7.6.1 Fundamental Differences between AC and DC-Type Motors

There are some fundamental differences between AC- and DC-type motors that dictate their power range, control, and applications. In the following sections, we will discuss these differences and their effects.

The first major difference between these motors is whether or not their speed can be controlled. This will be discussed later with servomotors, but suffice it to say that the speed of rotation of a DC motor can be controlled by changing the current to the windings; as the current increases or decreases, for the same load on the rotor, its speed also increases or decreases. However, the speed of an AC motor, among other things, is a function of the frequency of the supplied AC power. Since the frequency of the AC power is constant, the speed of an AC motor is generally constant.

The second major factor in the difference between brushed and brushless motors is the life of the brushes and commutators as well as the physical limitation of mechanical switching by brushes. Brushless DC motors, AC motors, and stepper motors are all brushless; therefore, they are sturdy and generally have a long life (only limited by the life of rotor bearings). Since brushes wear out, the life of brushed motors is limited and they require more maintenance.

The third important issue in the design and operation of all motors is heat dissipation. Like many other devices, the generated heat in motors eventually becomes the deciding factor about its size and power. The heat is generated primarily from the resistance of the wiring to electric current (load related), but includes heat due to iron losses including eddy current losses and hysteresis losses, friction losses, brush losses, and short-out circuit losses (speed related). As we will see in more detail later, the generated heat is a function of current ($W = I^2 R$). Thicker wires generate less heat but are more expensive, heavier (more inertia), and require more space. All motors generate this heat. However, what is important is the rate at which the heat is dissipated, and therefore, the path that the heat must take to leave the motor. The heat path is more important than the amount of heat, since if the dissipation is faster, more generated heat can be dissipated before damage occurs.

Figure 7.15 shows the heat dissipation path to the environment for an AC type motor and a DC-type motor. In DC-type motors, the rotor contains the winding and carries the

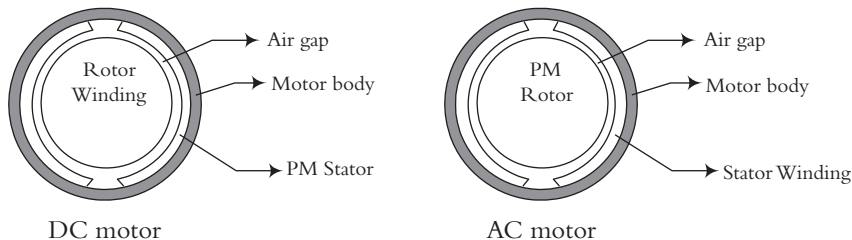


Figure 7.15 Heat dissipation path of motors.

current and, consequently, the heat is generated in the rotor. This heat must travel from the rotor, through the air gap, through the permanent magnets, through the motor's body, and be dissipated into the environment (it may also go through the shaft to the bearings and out). As you know, air is a good insulator. Therefore, the total heat transfer coefficient for the DC motor is relatively low. On the other hand, in an AC-type motor, the rotor is a permanent magnet and the winding is in the stator. The generated heat in the stator is dissipated to the air by conduction through the motor's body. As a result, the total heat transfer coefficient is relatively high, especially because no air gap exists. As a result, AC-type motors can be exposed to relatively higher currents without damage and, as a result, they are generally more powerful for the same size. Stepper motors and brushless DC motors, although not AC motors, have similar construction; the rotor is a permanent magnet and the stator contains the windings. Consequently, they also have better heat dissipation capability.

Overheating may be the most common cause of failure for electric motors. It can lead to:

- Failure of the winding isolation, causing shorts or burnouts,
- Failure of the bearings, resulting in jamming the rotor shaft,
- Degradation of the magnets, permanently reducing the motor's torque.

Heat development in a motor is not linear and is a function of the following parameters:³

- The basic heat generated in the winding.
- As the winding heats up, the resistance of the wiring increases, further increasing heat generation.
- The strength of the magnetic flux is negatively affected by heat, reducing the torque and requiring additional current.
- Heat dissipation increases as temperature rises.

Therefore, it is possible that the motor may achieve equilibrium at an acceptable temperature. However, there is also a danger that the temperature may continue to rise to failure. The heat generated in a motor is:

$$P_{electric} = i^2 R = \frac{T^2}{K_t^2} R_t \quad (7.10)$$

where i is the current in the winding, R and R_t are the nominal electrical resistance of the winding and the resistance at temperature t , T is the torque, K_t is the *torque constant* at temperature t and is a function of the magnetic field, number of turns of the windings, the effective area of the air gap, the radius of the rotor, and material properties. The variations in the winding electrical resistance can be described by:

$$R_t = R_{ref} [1 + (t_{winding} - t_{ref}) \cdot \alpha] \quad (7.11)$$

where R_t and R_{ref} are the electrical resistance at the temperature of interest $t_{winding}$ and room temperature t_{ref} , and α is a material constant. $\alpha_{copper} = 0.00393 \text{ (K}^{-1}\text{)}$. The torque constant, representing the magnetic field of the motor, varies with temperature as:

$$K_t = K_{ref} [1 + (t_{magnet} - t_{ref}) \cdot \beta] \quad (7.12)$$

Here, K_t and K_{ref} represent torque constants at the temperature of interest t_{magnet} and the reference temperature (20°C), and β represents the decay of magnetic flux-density, a material dependent property. Since this decay is negative, magnetic flux decreases as temperature increases. A simplified model for the final temperature of the motor can be written as:

$$t_{motor} = (P_{electric} + P_{friction})R_{thermal} + t_{ref} \quad (7.13)$$

where $R_{thermal}$ is the thermal resistance between the motor and the ambient. Through these equations, we can estimate the elevated temperature of the motor and find out whether or not it is at equilibrium.

Example 7.2

A motor develops a torque of 1.2 Nm at its nominal speed, with reference electrical resistance of 8Ω . The reference torque constant for the motor is 0.5 Nm/A and the thermal resistance for the motor is 1.05 K/W. Ambient temperature is assumed to be 20°C . For the material used in the motor, $\beta = -0.002$ (K). Friction is ignored.

- (a) Find whether or not the motor temperature will stabilize, and if so, find the converged value.
- (b) Repeat the same assuming that due to improvements in heat dissipation (for example, by using a fan), thermal resistance is reduced to 1.02 K/W.

Solution:

- (a) Substituting these values in Equations (7.10) through (7.13), we get the following power based on initial resistance:

$$P_{electric} = \frac{T^2}{K_t^2} R_t = \frac{1.2^2}{0.5^2} (8) = 46 \text{ W}$$

$$\text{and } t_{motor} = (P_{electric} + P_{friction})R_{thermal} + t_{ref} = (46 + 0)(1.05) + 20 = 68^\circ\text{C}$$

At this temperature, both the resistance and the torque constant change and we get:

$$R_t = R_{ref} [1 + (t_{winding} - t_{ref}) \cdot \alpha] = 8[1 + (68 - 20)(0.00393)] = 9.5\Omega$$

$$K_t = K_{ref} [1 + (t_{magnet} - t_{ref}) \cdot \beta] = 0.5[1 + (68 - 20)(-0.002)] = 0.452 \text{ Nm/A}$$

Resubstituting these values into Equations (7.10) and (7.13), we get a new temperature of:

$$P_{electric} = \frac{1.2^2}{0.452^2} (9.5) = 67 \text{ W}$$

$$t_{motor} = (67 + 0)(1.05) + 20 = 90^\circ\text{C}$$

The next iterations of the same process show that the temperature continues to rise. At iteration #20, the temperature is 180 degrees and rising uncontrollably, indicating that the motor will overheat.

- (b) Repeating the same with the new thermal resistance will converge after about 30 iterations to near 130°C . Therefore, it is clear that the proper dissipation of heat is a major issue in motors. ■

It should be clear from the preceding discussion that AC-type operation is preferable when considering heat generation and brushless operation, whereas DC-type operation is preferred for speed control or when only DC power is available. A combination of both characteristics would be ideal. As we will see in more detail later, brushless DC motors and stepper motors possess these characteristics, and therefore, are more robust and last longer.

In the following sections, we will briefly discuss DC motors, servomotors, brushless DC motors, and stepper motors.

7.6.2 DC Motors

DC motors are very common in industry, and have been used for a long time. As a result, they are reliable, sturdy, and relatively powerful.

In DC motors, the stator is a set of fixed permanent magnets, creating a fixed magnetic field, while the rotor carries a current. Through brushes and commutators, the direction of the current is changed continuously, causing the rotor to rotate continuously. Conversely, if the rotor is rotated within the magnetic field, a DC current will develop, and the motor will act as a generator (the output is DC, but not constant). Figure 7.16 shows the construction of a DC motor, its commutators and brushes, and the permanent magnet stator.

If permanent magnets are used to generate the magnetic field, the output torque T is proportional to the magnetic flux ϕ and the current in the rotor windings i . Then:

$$T = \alpha \cdot \phi \cdot i = k_t \cdot i \quad (7.14)$$

where k_t is called the *torque constant*. Since in permanent magnets, the flux is constant, the output torque becomes a function of i , and to control the output torque, i (or corresponding voltage) must be changed. If, instead of permanent magnets, soft iron cores with windings are used for the stator as well, then the output torque is a function of currents in both the rotor and the stator windings, as:

$$T = k_t k_f i_{\text{rotor}} i_{\text{stator}} \quad (7.15)$$

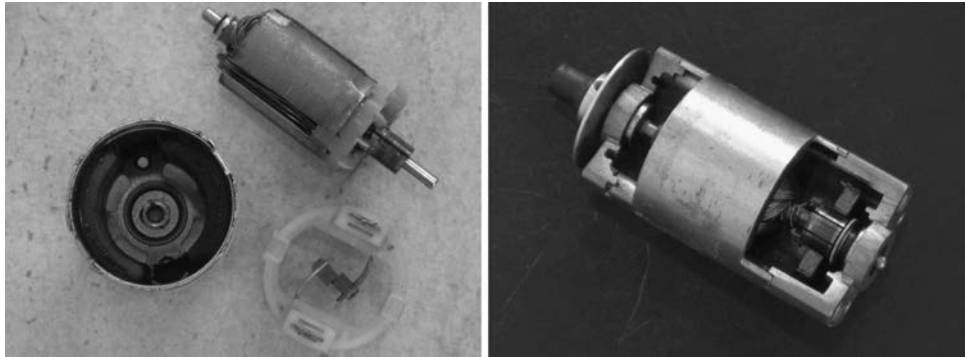


Figure 7.16 The stator, rotor, commutators, and brushes of a DC motor. (© Saeed Niku.)

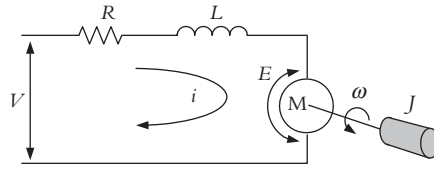


Figure 7.17 Schematic diagram showing a DC motor armature circuit.

where both k_t and k_f are constants. Assuming no power loss during this energy conversion, the total input must be equal to the output. Therefore:

$$P = T \cdot \omega = E \cdot i \rightarrow E = \frac{T \cdot \omega}{i} = k_t \cdot \omega \quad (7.16)$$

which indicates that voltage E is proportional to the angular velocity of the motor ω . This voltage is called the back-emf voltage and is generated across the motor because the windings cross the magnetic field. Therefore, as the rotor speed increases, so does the back-emf voltage. Since in reality the rotor windings have both resistance and inductance, we can write (see Figure 7.17):

$$V = Ri + L \frac{di}{dt} + E \quad (7.17)$$

Substituting Equations (7.14) and (7.16) into (7.17) and rearranging it, we get:

$$\frac{k_t}{R} V = T + \frac{L}{R} \frac{dT}{dt} + \frac{k_t^2}{R} \omega \quad (7.18)$$

$\frac{L}{R}$ is called *motor reactance* and is usually small, and therefore, the differential term may be ignored for simplicity of analysis at this time. Consequently, we get:

$$T = \frac{k_t}{R} V - \frac{k_t^2}{R} \omega \quad (7.19)$$

Equation (7.19) shows that as input voltage V increases, the output torque of the motor increases as well. It also shows that as the angular velocity increases, the torque decreases due to back-emf. Therefore, when $\omega = 0$, torque is the greatest (stalled motor), and when ω is at its nominal maximum value, $T = 0$ and the motor does not produce any useful torque (see Figure 7.18).

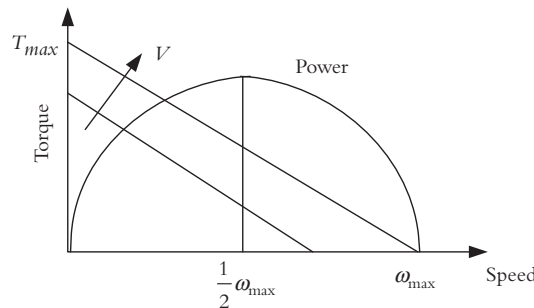


Figure 7.18 The output torque and power of a DC motor versus its angular velocity.

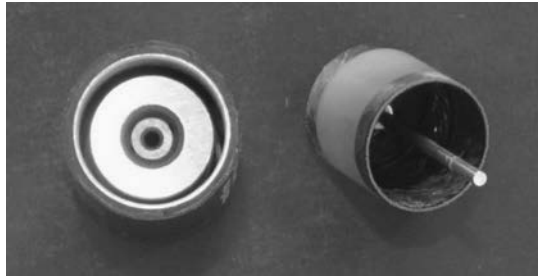


Figure 7.19 Eliminating the soft iron from a rotor makes it lightweight with low inertia. (© Saeed Niku.)

Referring to Equation (7.16), the output power of the motor is zero when either the angular velocity of the motor (at stall condition) or the torque (at maximum speed) are zero (see Figure 7.18).

Through the use of powerful magnets made of rare-earth materials and alloys such as neodymium, the performance of motors has improved significantly. As a result, the power to weight ratio of motors is much better than before, and they have replaced almost all other types of actuators. However, these motors are also more expensive.

To overcome the problem of high inertia and the large size of many electric motors, a disk or hollow rotor can be used. In these motors, the iron core of the rotor winding is eliminated to reduce its weight and inertia; as a result, these motors are capable of producing very large accelerations (zero to 2000 rpm in one ms^5), and they respond very favorably to changing currents for control purposes. Figure 7.19 shows a motor in which the hollow rotor has no soft iron. In a disk motor, the rotor is a flat, thin plate with windings pressed (etched) into it, as if we would flatten a rotor into a disk. The permanent magnets are generally small, short, cylindrical magnets that are placed on the two sides of the disk. As a result, disk motors are very thin and are used in many applications where both space and acceleration requirements are important (Figure 7.20).

7.6.3 AC Motors

Electric AC motors are similar to DC motors, except that the rotor is permanent magnet, the stator houses the windings, and all commutators and brushes are eliminated. This is possible because the changing flux is provided by the AC current, and not by

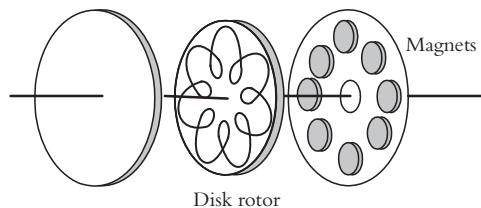


Figure 7.20 Schematic of a disk motor. The rotor has no iron core and consequently, has very little inertia. As a result, it can accelerate and decelerate very quickly.



Figure 7.21 A center-tapped AC motor winding.

commutation. As the flux generated by the AC current changes, the rotor follows it and rotates. As a result, AC motors have fixed nominal speeds, a function of the number of poles on their rotor, and the line frequency (e.g., 60 Hz). Since AC motors can dissipate heat more favorably than DC motors, they can be more powerful. The same principles of back-emf (see section 7.6.6) hold for AC motors as well.

Nowadays, with the available power electronics, it is possible to use methodologies such as *flux vector control* to generate a sinusoidal current with desired frequency and amplitude that enable us to control the speed and torque of an AC motor. This is accomplished by converting the AC line voltage to a DC form and creating an approximate pulsed sinusoidal DC current at the desired frequency and magnitude to drive the motor. Stepper motors (driven in microstepping mode) and brushless DC motors are similar attempts at making DC motors operate similar to an AC motor.

There are also reversible AC motors available. In this case, the motor winding is center-tapped; therefore, as the current flows in each half of the winding, the direction of the flux, and consequently the rotation of the rotor, changes (Figure 7.21). However, since the current flows in only half of the winding, the generated torque is half as much.

7.6.4 Brushless DC Motors

Brushless DC motors are a hybrid between AC motors and DC motors. Although not exactly the same, their construction is very similar to an AC motor. The major difference is that brushless DC motors are operated with an electronically switched DC waveform that is similar to an AC current (either sine wave or trapezoidal waveform) but is not necessarily at 60 Hz. As a result, unlike AC motors, DC brushless motors can be operated at any speeds, including very low speeds. To operate, a feedback signal is necessary to determine when to switch the direction of the current. In practice, a resolver, an optical encoder, or Hall-effect sensors send signals to a controller, which switches the current to the stator. For smooth operation and almost constant torque, the stator usually has three phases in it.^{2,4} Therefore, three currents, with 120° phase shift, are fed into the stator. Brushless DC motors are operated by a controller circuit. They will not operate if you connect them directly to a DC power source. Figure 7.22(a) shows a radial-type brushless DC motor with the rotor and stator side by side. Figure 7.22(b) is the stator of an axial-type brushless motor.

7.6.5 Direct Drive Electric Motors

Direct drive electric motors are very similar in construction to brushless DC motors or stepper motors. The major difference is that they are designed to deliver a very large torque at very low speeds, with very high resolution. These motors are intended to be used directly with a joint without any gear reduction. Direct drive motors can be very expensive and very heavy, but have impressive characteristics. In one model by

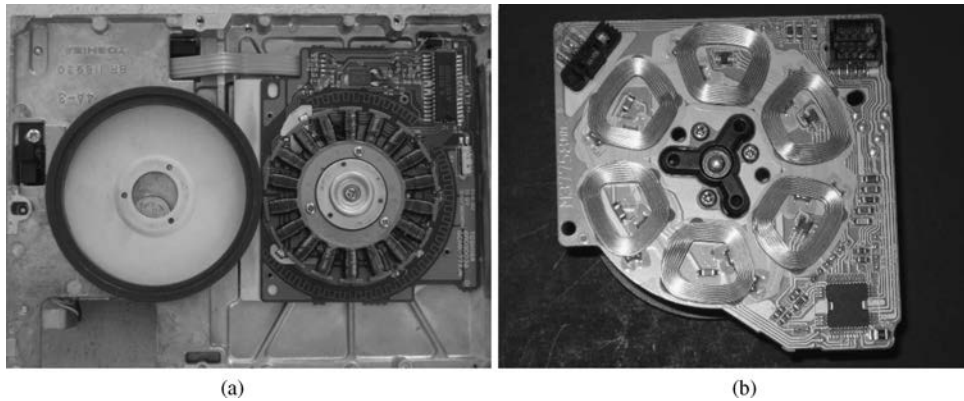


Figure 7.22 Brushless DC motors. (© Saeed Niku.)

NSKTM Corporation, a 40 Kg motor produces a continuous torque of 150 Nm at 3 rps maximum, with a resolution of 30 arc-sec.

A voice coil may be used as a direct drive actuator for low torque but high resolution applications. Voice coils are commonly used in disk drives and deliver impressive characteristics in reliability and resolution. Figure 7.23 shows a disk drive voice coil actuator.

7.6.6 Servomotors

An important issue in all electric motors is the back electromotive force, or back-emf. As discussed earlier, a conductor carrying a current within a magnetic field will experience a force, which causes it to move. Similarly, if a conductor moves within a magnetic field such that it crosses the field lines, a current will be induced into the conductor. This is the basic principle of electric power generation. However, it also means that when the wires of the windings in a motor are rotating within the magnetic field of the magnets, a current (or voltage) will be induced in them in the opposite direction of the input current, called



Figure 7.23 Disk drive voice coil actuator. (© Saeed Niku.)

back-emf, that tends to reduce the effective current of the motor. The faster the motor rotates, the larger the back-emf. Back-emf current is usually expressed as a function of rotor speed as shown in Equation (7.16) and repeated here:

$$E = k_t \cdot \omega \quad (7.16)$$

where k_t is typically given in volts per 1000 rpm. As the motor approaches its nominal no-load speed, the back-emf is large enough, such that the motor speed will stabilize at the nominal no-load speed with its corresponding effective current. However, at this nominal speed, the output torque of the motor is essentially zero. The motor's velocity is governed by Equation (7.19), repeated here:

$$T = \frac{k_t}{R} V - \frac{k_t^2}{R} \omega \quad (7.19)$$

As shown in Figure 7.18, at maximum ω , the output torque is zero. For constant input voltage V , if a load is applied to the motor, it will slow down, resulting in smaller back-emf, larger effective current, and consequently, a positive net torque. The larger the load, the slower the motor will rotate in order to develop a larger torque. If the load becomes increasingly larger, there comes a time when the motor stalls, there is no back-emf, the effective current is at its maximum, and the torque is at its maximum. Unfortunately, in each case, when the back-emf is smaller, although the output torque is larger, since the net current is larger, so is the generated heat. Under stall or near-stall conditions, the generated heat may be large enough to damage the motor.

To increase the motor torque while maintaining a desired speed, the input voltage V (or current) to the rotor, stator, or both if soft iron magnets are used, must be increased. In such a case, although the motor rotates at the same speed, and although the back-emf is still the same, the larger voltage will increase the net effective current, and consequently, the torque. By varying the voltage (or corresponding current), the speed-torque balance can be maintained as desired. This system is called a servomotor.

A servomotor is a DC, AC, brushless, or even stepper motor with feedback that can be controlled to move at a desired speed and torque for a desired angle of rotation. To do this, a feedback device sends signals to the servocontroller circuit reporting its angular position and velocity. If, as a result of higher loads, the velocity is lower than the desired value, the voltage (or current) is increased until the speed is equal to the desired value. If the speed signal shows that the velocity is larger than desired, the voltage is reduced accordingly.⁶ If position feedback is used as well, the feedback signal is used to shut off the motor as the rotor approaches the desired angular position.

We will discuss sensors later. Here, suffice it to say that many different types of sensors may be used for this purpose, including encoders, resolvers, potentiometers, and tachometers. If a position sensor such as a potentiometer or encoder is used, its signal can be differentiated to produce a velocity signal. Figure 7.24 is a schematic of a simple control block diagram for a servomotor. Refer to Chapter 6 for more detail.

7.6.7 Stepper Motors

Stepper motors are versatile, robust, simple motors that can be used in many applications. In most applications, stepper motors are used without feedback. This is because, unless a

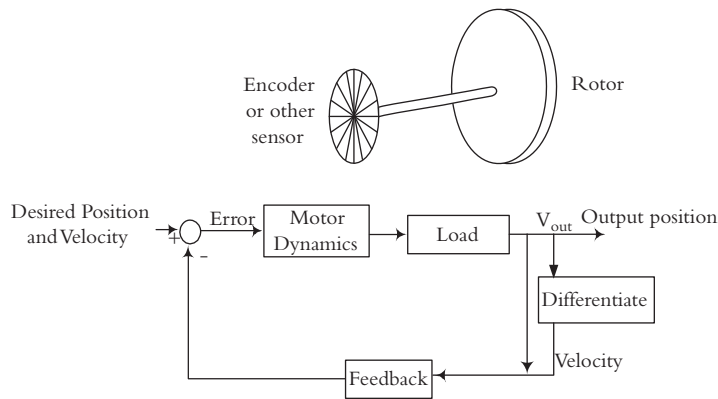


Figure 7.24 Schematic of a servomotor controller. A sensor sends velocity and position signals to the controller, which controls the output velocity and position of the servomotor.

step is missed, the motor steps a known angle each time it is moved. Therefore, its angular position is always known and no feedback is necessary. Stepper motors come in many different forms and principles of operation. Each type has certain characteristics unique to it, yielding it an appropriate choice for particular applications. Most stepper motors can be used in different modes by wiring them differently.

Unlike regular DC or AC motors (but like brushless DC motors) if you connect a stepper motor to power, it will not rotate. Steppers rotate only when the magnetic field is rotated through its different windings. In fact, their maximum torque is developed when they do not turn. Even when not powered, steppers have a residual torque called *detent* or *residual torque*. An external torque must be applied to turn a stepper motor even when not powered. Stepper motors need a microprocessor or driver/controller (indexer) circuit for rotation. You may either create your own driver, or you may purchase a device called an indexer that drives the stepper motor for you. Similar to servomotors, which need feedback circuitry, stepper motors need drive circuitry. So, in each application, the designer must decide which type of motor is more appropriate. For industrial robotic actuation, except in small tabletop robots, stepper motors are hardly ever used. However, stepper motors are used extensively in nonindustrial robots and robotic devices as well as in machines used in conjunction with robots, from material handling machines to peripheral devices and from automatic manufacturing to control devices.

Structure of Stepper Motors Generally, stepper motors have soft iron or permanent magnet rotors, while their stators house multiple windings. Based on the discussion in section 7.6.1, since the heat generated in the coils can more easily dissipate through the motor's body, stepper motors are less susceptible to heat damage, and since there are no brushes or commutators, they have long life.

Rotors of stepper motors are not all alike. We will discuss two types of rotors later. In each case though, the rotor follows a moving magnetic field generated by the coils. As a result, somewhat similar to both AC motors and brushless DC motors, a rotor follows a moving flux under the control of a controller or driver. In the next sections, we will study how stepper motors operate.

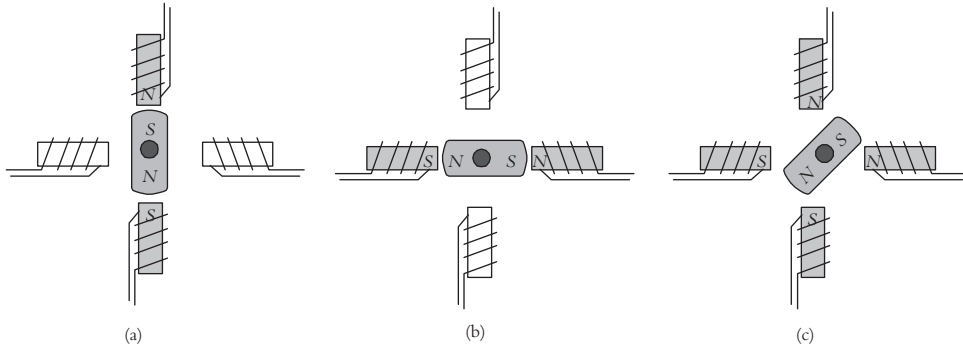


Figure 7.25 Basic principle of operation of a stepper motor. As the coils in the stator are turned on and off, the rotor will rotate to align itself with the magnetic field.

Principle of Operation Generally, there are three types of stepper motors: variable reluctance, permanent magnet (also called *canstack*), and hybrid. Variable reluctance motors use a soft iron, toothed, rotor. Permanent magnet canstack-type motors use a permanent magnet rotor without teeth. Hybrid motors have a permanent magnet, but toothed, rotor. These simple differences result in somewhat different principles of operation. In the following sections, we will discuss canstack- and hybrid-type motors. The variable reluctance motors operate with the same principle as hybrid motors.

Imagine a stepper motor with two coils in its stator and a permanent magnet as its rotor, as in Figure 7.25. When one of the coils of the stator is energized, the permanently magnetized rotor (or soft iron in variable reluctance motors) will rotate to align itself with the stator magnetic field (a). The rotor will stay at this position unless the field rotates. As the power to the present coil is discontinued and is directed to the next coil, the rotor will rotate to once again align itself with the field in the new position (b). Each rotation is equal to the step angle, which may vary from 180 degrees to as little as a fraction of a degree (in this example, 90°). Next, the first coil will once again be turned on, but in the opposite polarity, while the second is turned off. This will cause the rotor to rotate another step in the same direction. The process continues as one coil is turned off and another is turned on. A sequence of four steps will bring the rotor back to exactly the same state it was at the beginning of the sequence.

Now imagine that at the conclusion of the first step, instead of turning off one coil and turning on the second coil, both would be turned on. In that case, the rotor would only rotate 45° to align itself to the path of least reluctance (c). Later, if the first coil is turned off while the second remains on, the rotor will rotate another 45° . This is called half-step operation and includes a sequence of eight movements.

Of course, with the opposite on-off sequence, the rotor will rotate in the opposite direction. Typical steppers run between 1.8 to 7.5 degrees in full-step mode. Obviously, one way to reduce the size of the steps is to increase the number of coils. However, there is a physical limit to how many coils may be used. To further increase the number of steps per revolution, different numbers of teeth can be built into the stator and rotor creating an effect similar to a caliper. For instance, 50 teeth on the rotor and 40 teeth on the stator will result in a 1.8 degree step angle with 200 steps per revolution, as will be discussed later.

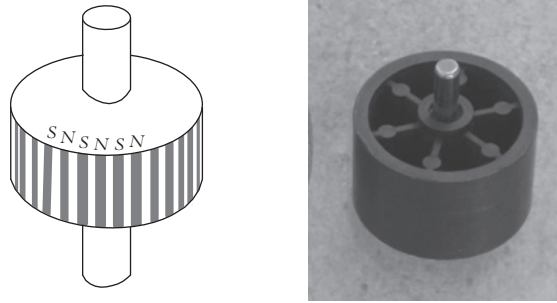


Figure 7.26 The rotor of a canstack stepper motor. (© Saeed Niku.)

Canstack Motors These motors are very common, are usually 7.5° steppers or similar, and are used in many different applications. Due to their construction, they are relatively short and lend themselves to applications with low vertical clearance.

The rotor is a cylinder with alternate strips of north-south polarities, usually made of resins embedded with ferrite particles (similar to refrigerator magnets), as shown in Figure 7.26. A typical rotor for a 7.5° stepper motor will have 24 pairs of poles on it.

A refrigerator magnet sticks to ferrite materials on one side only; the side that is meant to have a message or advertisement on it does not stick. Is it because of the additional printed layer? Not really. It is because there is no magnetic flux on that side, but only on the side that sticks. These magnets are made of a resin mixed with a ferrite powder that is magnetized as a series of small horseshoe magnets next to each other called *Halback Array*, as shown in Figure 7.27.

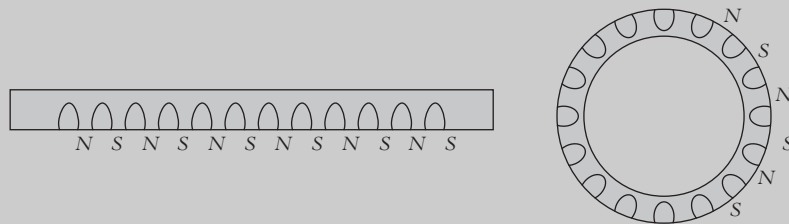


Figure 7.27 Magnetic flux pattern on a refrigerator magnet and the rotor of a stepper motor or brushless DC motor.

The permanent magnet rotors of many stepper motors and brushless DC motors are made the same way, and therefore, the rotor's cross-section is made up of a series of horseshoe magnets next to each other. This arrangement creates a unique rotor that allows small steps in a brushless DC motors (see Figure 7.22) and canstack-type stepper motor (see Figure 7.26).



Figure 7.28 The stator of a canstack stepper motor. (© Saeed Niku.)

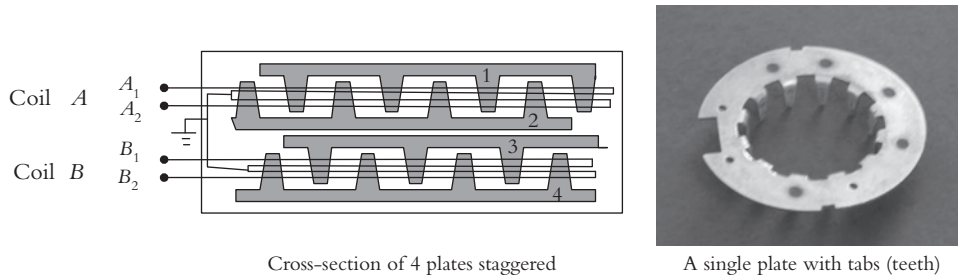


Figure 7.29 Canstack stator windings and plates. (© Saeed Niku.)

The stator is made up of four plates, each with 12 teeth, stacked on top of each other, with the teeth staggered in 1-3-2-4 order as shown in Figures 7.28 and 7.29. One coil is wrapped around plates 1 and 2, as is one around plates 3 and 4. The four plates create two independent magnets on top of each other, each with a winding that is center-tapped (called bifilar) and grounded at the center. A current going through one coil and exiting at the center will create a certain magnetic polarity which is the opposite of the polarity if the current goes in from the other end and exits at the center wire. As a result, each coil may be energized at either polarity. Each winding will cause all the teeth on each plate of a pair to be of similar polarity: all north or all south. Consequently, energizing both windings will create repeating patterns of north and south in all four plates.

Figure 7.29 is a schematic of a canstack stator, linearized for better visualization. Plates 1 and 2 are related to coil A_1/A_2 (two halves of coil A), as are plates 3 and 4 to coil B_1/B_2 (two halves of coil B). If coil A_1 is turned on, plate 1 will be north and plate 2 will be south. However, if coil A_2 is turned on, the polarity of plates 1 and 2 will be reversed; plate 1 will be south and plate 2 will be north. The same will happen to plates 3 and 4 with coils B_1 and B_2 . This is called center-tapping and is shown in Figure 7.30. The advantage

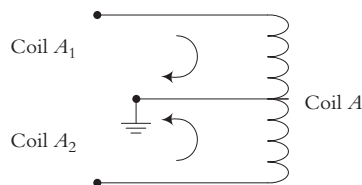


Figure 7.30 Center tapping of a coil allows changing the polarity of the magnetic field by having the current flow in either half of the coil.

of center tapping is that during manufacturing, a set of two wires is simultaneously wrapped around the pole. At the conclusion, the two heads of the wires are connected and form the ground, while the other two heads will be contact points for input current. As a result, the magnet may easily be energized with either polarity by simply having current in each half of the double winding.

During the operation of the motor, two coils will be turned on in the following sequence resulting in the indicated polarities:

Step	A1	A2	B1	B2	Plate1	Plate2	Plate 3	Plate 4
1.	on	off	on	off	N	S	N	S
2.	off	on	on	off	S	N	N	S
3.	off	on	off	on	S	N	S	N
4.	on	off	off	on	N	S	S	N

Notice that A1 and A2 or B1 and B2 are never turned on simultaneously (as they would cancel each other's fields). During each step of the sequence, the poles on the rotor will align themselves between the stator poles with the least reluctance such that any south pole on the rotor will be between two north poles on the stator, and any north pole on the rotor will be between two south poles on the stator, as in Figure 7.31. In this Figure, “arcs” of S-N show the polarities of the poles at each step of the sequence. At the end of the four-step sequence, the rotor has moved four steps, which will bring it to exactly the same situation as in the beginning of the sequence. Therefore, repeating the four-step sequence will rotate the rotor continuously. The faster the sequencing of steps, the faster the rotor will rotate. As a result, by carefully controlling how many sequences

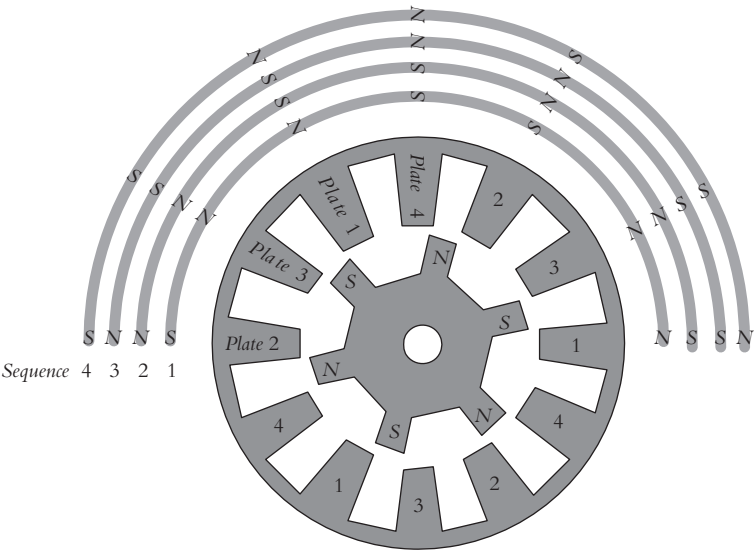


Figure 7.31 The cross-section of a canstack stepper motor. Each S-N arc shows the polarities of each pole during one step of the sequence of four.

Table 7.2 (a) Full Step and (b) Half Step Sequence for Stepper Motors.

(a)					(b)				
Step	A1	A2	B1	B2	Step	A1	A2	B1	B2
1	On	Off	On	Off	1	On	Off	On	Off
2	Off	On	On	Off	2	Off	Off	On	Off
3	Off	On	Off	On	3	Off	On	On	Off
4	On	Off	Off	On	4	Off	On	Off	Off
					5	Off	On	Off	On
					6	Off	Off	Off	On
					7	On	Off	Off	On
					8	On	Off	Off	Off

are provided and at what speed, the rotational displacement as well as angular velocity of the motor can be controlled. Note that Figure 7.31 is drawn with only 1/4 of all stator and rotor poles. In actual stepper motors, there are a total of 48 poles, providing a 7.5° step angle.

Instead of always energizing two coils simultaneously, if either one or two coils are energized alternately, rendering an eight-step sequence, the stepper will step at half the angle, therefore halfstepping. However, this is not common in canstack motors. Table 7.2 shows the on-off sequence for full stepping as well as for halfstepping the stepper motor. Reversing the sequence will cause the stepper motor to rotate in the opposite direction.

Hybrid Stepper Motors These steppers are usually made with two coils—either center tapped, or two independent windings in opposite directions—each with four poles. The rotor is made of two collinear cylinders mounted on a stainless steel shaft, such that one end of the rotor is north and the other end is south (Figures 7.32 and 7.33). The rotor and the stator poles are all cut to have teeth, where the teeth on one half of the rotor are offset by a half tooth from the other half of the rotor. However, to understand the role

**Figure 7.32** Stator of a hybrid stepper motor. (© Saeed Niku.)

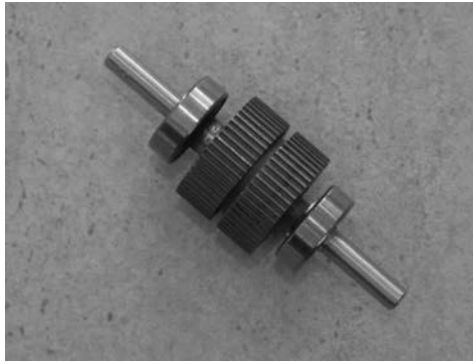


Figure 7.33 Rotor of a hybrid stepper motor. (© Saeed Niku.)

of the teeth on the rotor and stator of these motors, we first review the concept behind a caliper.

Let's take two parallel lines capable of sliding relative to each other, each one unit of length long. Next, divide each line into 10 equal divisions. In order to have the division lines aligned with each other when moving one step, the sliding line will have to move one whole division to the next line (Figure 7.34).

Similarly, take two other lines, divide one into ten equal divisions and the other line into 11 equal divisions. In this case, the divisions will be 0.1 and approximately 0.09, respectively. If two of the division lines are aligned and one of the lines slides, it will only take a distance of $0.1 - 0.09 = 0.01$ units to align the next pair of division lines. Calipers do exactly the same. By dividing the lines into different lengths, we can easily measure a distance at a fraction of the smallest division.

The teeth on the stepper motor stator and rotor are also the same. Since the number of teeth on the stator and rotor are different, for each step, the rotor only rotates an angle equal to the difference between the two divisions. With 40 teeth on the stator poles and 50 teeth on the rotor, the step angle will be 1.8 degrees at full steps, because:

$$\frac{360^\circ}{40} - \frac{360^\circ}{50} = 9^\circ - 7.2^\circ = 1.8^\circ \quad (7.20)$$



Figure 7.34 Application of unequal divisions for measuring lengths as in a caliper.

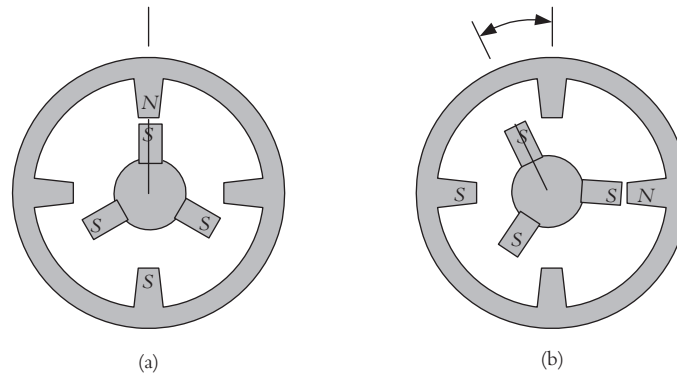


Figure 7.35 Basic operation of a hybrid stepper motor.

The two rotor cylinders (Figure 7.33) are a half-step out of phase and the stator poles run across the whole length of the stepper. Although the construction of hybrid and canstack steppers are different, driving the motors is similar.

Figure 7.35 is a simplified hybrid stepper motor with only two coils and a rotor with three teeth. As we discussed earlier, in actual motors, many teeth are cut into the poles to create smaller steps, but to see how the motor works this example only uses three teeth. As long as the number of teeth in the stator and the rotor are different, the above-mentioned effect will be true. Now suppose that one coil is energized as shown in (a). The rotor, with all its teeth as south on one side and north on the other will align itself to the path of least reluctance, as shown. If the coil is turned off and the second coil is turned on, although the flux (or field) has turned 90° , the rotor will rotate only 30° to the new location in (b). The same sequence will continue similarly to a canstack motor and the rotor will rotate accordingly. Changing the sequence backwards will cause the rotor to rotate backwards. Applying the sequence faster will cause the rotor to move faster. Therefore, by controlling the sequence, its direction, and its speed, the rotor's motion and angular velocity can be controlled. Of course, the same eight-step sequence can also be applied to these motors for half-step operation.

Unipolar, Bipolar, and Bifilar Stepper Motors Unipolar stepper motors are designed to work with one power source. Generally, it is desirable to have one power source that powers both the motor windings as well as the drive circuits. Unless other techniques of switching are used, with one power source, it is impossible to simply change the polarity of the coils by changing the polarity of the power from the power source as this would ruin the electronic drive circuitry. Since the polarity of the coils cannot be changed, these motors may not be run in half-step mode. However, there is only one power source used, which reduces the cost, and the motor will develop its full power.

For bipolar motors, the assumption is that the polarity of the power source can be changed. As a result, there will either be two power sources, where one powers the motor windings and its polarity can be switched, and one is used to power the drive circuitry, or that more sophisticated switching is used with one power supply to prevent damage to the circuits. In this case, the motor will require more power supplies or

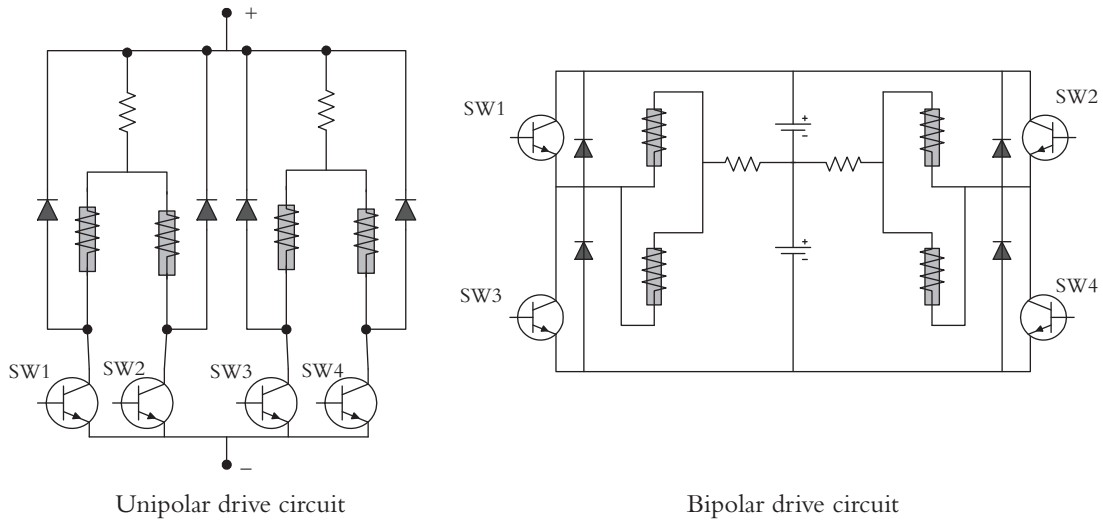


Figure 7.36 Schematic drawing for unipolar and bipolar drive circuits.

electronics, but can be run either at full or half-step modes, and will develop its full power.

In bifilar motors, the coils are center-tapped, as was previously discussed. In this case, the polarity of the coils can be changed simply by flowing the current in each half of the coil. Therefore, with simple circuits and one power supply, the motor can be run in either full or half-step mode, but since only half of the coil is energized, the motor develops only half of its full power. Figure 7.36 is a schematic drawing for unipolar and bipolar drive circuits. The switches are connected to the microprocessor ports and are turned on and off by the microprocessor.

Most motors are wired to be used in any of the three modes. Some motors are not. Figure 7.37 shows how stepper motors may be wired. In 8-lead configuration, you may wire the motor in any mode, as the coils are completely detached from each other. In this case, it is also easy to find which two wires are connected to which coil. The 6-lead motors are connected such that each coil is center tapped, but the two coils are detached. In 5-lead motors, the two center taps are connected together. 4-lead motors may not be used in bifilar mode. By measuring the resistance between different wires, we may find which wire is connected to which coil.

Figure 7.38 shows how a 5-phase “pentagon” stepper motor may be connected.

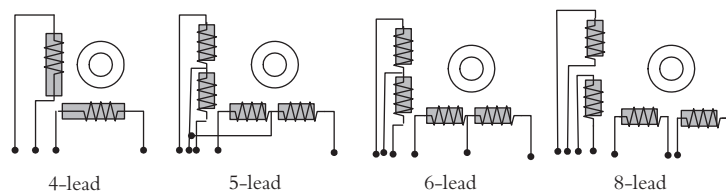


Figure 7.37 Stepper motor lead configurations.

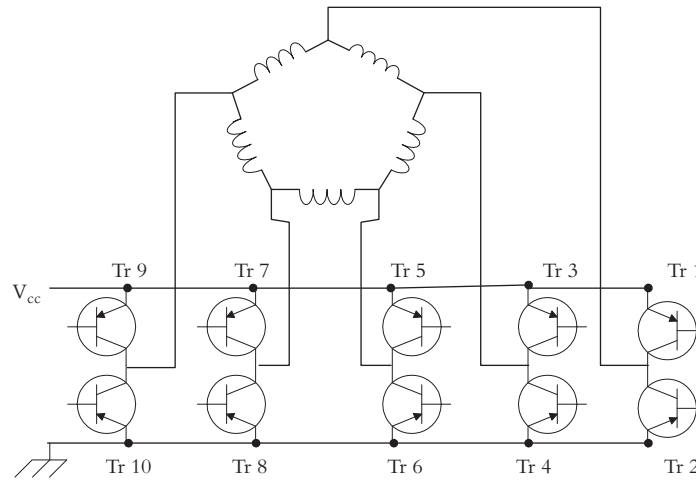


Figure 7.38 Connection schematic for a bipolar drive mode with pentagon connection for a 5-phase stepper motor. (Adapted by permission from the Oriental Motor USA Corp.)

Stepper Motor Speed-Torque Characteristics Stepper motors are very useful in many applications. They do not require any feedback, as it is assumed that stepper motors advance a known angle every time a signal is sent. As long as the load on the motor is less than the torque it can deliver, the steps will not be missed, and this assumption is correct. However, if the load is too large, or in fact, if the speed is more than what the motor is capable of rotating, steps may be missed, and since there is no feedback, all subsequent positions will be incorrect.

Stepper motors develop their maximum torque, called *holding torque*, at zero angular velocity, when the rotor is stationary (the torque developed with no power is called *detent* or *residual torque*). Like all motors, as the speed of the stepper motor increases, the torque it develops reduces, but more significantly than others. Therefore, it is very important that the designer checks the speed-torque characteristic of stepper motors from manufacturers before a choice is made. Figure 7.39 is a typical speed-torque characteristic. The useful torque (also called *pull-out torque*) depends on the way the stepper is externally wired and the drive power signals used.

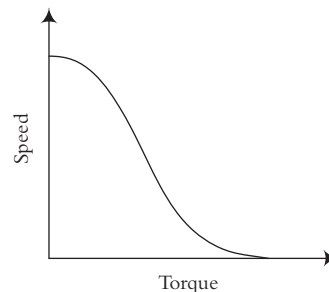


Figure 7.39 A typical speed-torque curve for a stepper motor.

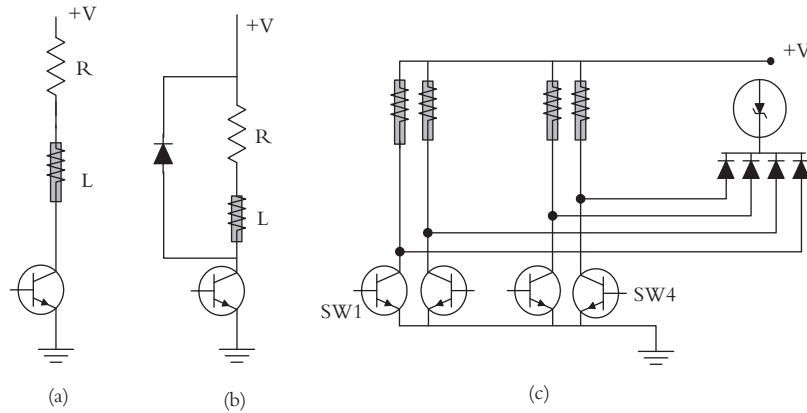


Figure 7.40 Application of (a) a resistor, (b) a diode, and (c) a zener diode with stepper motors for increasing the maximum velocity.

One reason for this poor performance at higher speeds is that since at each step, the rotor must accelerate, cruise, decelerate, and stop at the next step, and that this process must be repeated for every single step, stepper motors cannot rotate quickly, especially if the rotor is heavy. If the signals coming to the motor are too fast, the rotor will not have the time to accelerate/decelerate and will miss steps. So, one reason for losing output torque at higher speeds is the inertia load. However, even more important is the alternating magnetic field of the stator. As was discussed earlier, every time a coil is turned off, there will be a varying (in this case a decaying) flux. The wires of the coils, in the presence of a changing flux, will develop a back-emf that will slow down the decay of the flux as a new one builds up due to the back-emf current. The generated flux will tend to “hold” the rotor from rotating and will slow it down. As a result, the rotor is pulled back and is not able to rotate freely.

To remedy this problem and to prevent high-current “sparking” across the switches, a freewheeling diode may be added to the circuit, as in Figure 7.40(b). The diode will allow the current to continue flowing through the coil and be dissipated through conversion to heat in the resistor. The effectiveness of the process can be increased by adding a zener diode to the circuit, as shown in Figure 7.40(c). It is important to realize that the zener diode’s breakdown voltage must be near the transistor’s breakdown voltage rating. Otherwise, it will have no effect.

Microstepping Stepper Motors In microstepping, instead of turning the coils on and off abruptly, the power-up or power-down for each coil is done gradually by dividing the changes into smaller divisions, usually up to 250 steps. As an example, suppose coil *A* is on and coil *B* is off. In full stepping, the next sequence will be for coil *A* to be off and for coil *B* to be on. In microstepping, this is divided, let’s say, into 100 divisions. As a result, in the next microstep, coil *A* will be 99% on and coil *B* will be 1% on. Therefore, the rotor will turn slightly to the point of least reluctance, which is a microstep away from the previous original step. In the next step, coil *A* would become 98% strong and coil *B* would be at 2%, still microstepping the rotor a little further. The process continues until both are equal for the half-step, and then, until coil *A* is turned off

and coil *B* is 100% on. This would divide the full step into 100 smaller steps. For a 200-step 1.8° stepper motor, this means that with microstepping, the motor will have 20,000 steps per revolution. In practice, each step is usually divided into 125 or 250 microsteps, requiring 25,000 or 50,000 microsteps per revolution for a 1.8° stepper.

This dividing of the voltages is done with electronic circuits that resolve the voltage into smaller divisions and corresponding steps. Since, unlike full-stepping, the dissipation of the flux is not abrupt, the back-emf is much smaller. As a result, with microstepping, stepper motors' performance is improved significantly. The motor develops a higher torque, and its maximum speed without missed steps is much higher. Additionally, because microsteps are very small, the motor's movement is not as piecewise and, consequently, vibration of the motor is smaller and less pronounced. Of course, the disadvantage is that microstepping requires a more sophisticated driver that is much more accurate with better current resolution, and is more expensive.

In practice, instead of dividing the steps into linear and equally divided divisions (like 1% divisions), the changes follow a sine wave. In other words, the voltage to each coil is a piecewise, digital voltage resolved into up to 250 steps. Then, principally, in this mode, the stepper is in fact similar to an AC synchronous motor, except that its angular velocity is not fixed with line frequency but is driven by the microstepper driver circuit. As a result, the motor is driven at a desired angular velocity, for any desired angular displacement, and can be stopped at any instant.

Stepper Motor Control Stepper motors may be driven by microprocessors (or microcontrollers) either directly or through driver circuits. They can also be driven by a dedicated stepper driver/translator that motor manufacturers offer.

To drive a stepper motor directly with a microprocessor, the power to the motor coils must be directly controlled by the processor. This is accomplished in two ways, depending on the microprocessor. If the output port of the processor is low power (mA), it will not be able to provide a high enough current to the motor windings (the output port of a personal computer is an example of this situation). As a result, the output port must turn on and off power transistors that control the power flow into the motor coils at higher currents. If the output ports of the microprocessor can provide high current flow, so long as the current requirement of the motor is below the level of the microprocessor's, the motor coils can be directly connected to the output ports. In either case, the microprocessor controls the current flow to the coils of the stepper motor by sequentially turning on and off the output ports, as discussed previously. The stepper will move in one direction or another depending on the order of the sequence. In this situation, four output ports are needed to drive one stepper motor, but the stepper's displacement, velocity, and direction are all under control. Figure 7.41 shows a schematic arrangement in which a microprocessor with a low power output port is used to drive a stepper motor with transistors (a) as well as a microprocessor with high current capability, which is used without transistors (b).

Alternatively, it is possible to use a dedicated IC chip to control the motion of a stepper motor.^{7,8} These IC chips, called stepper driver with translator, are designed to sequence the stepper motor based on the information they receive. In most cases, the information needed is a stream of pulses. Every time the driver receives a pulse (the input to the chip changes from low to high), it will sequence the stepper by one step. If "n" signals are received, the motor will be sequenced for "n" steps. When a driver is used, the

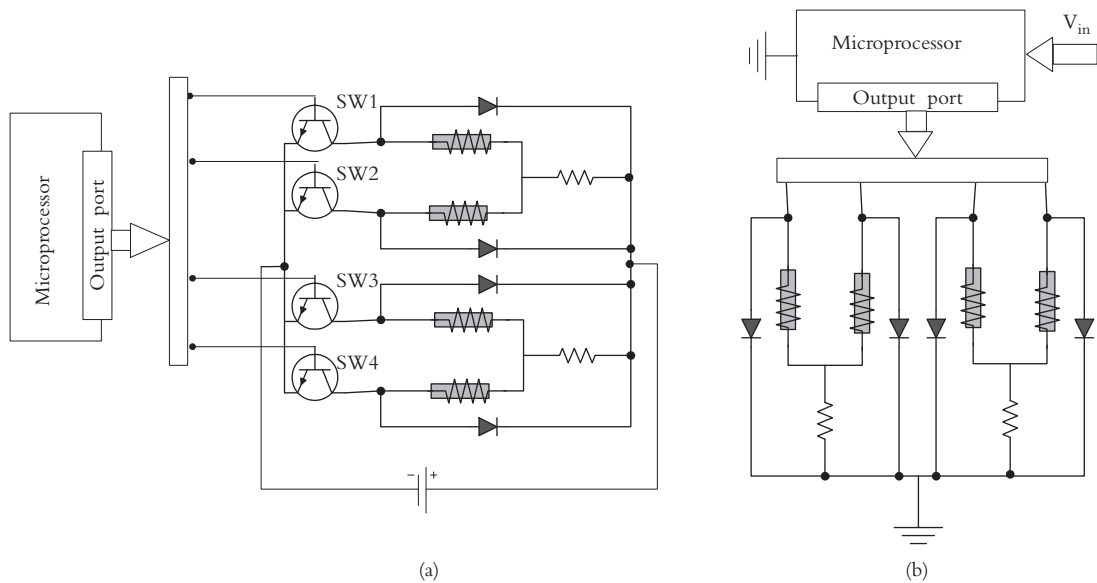


Figure 7.41 Schematic drawing of the application of a microcontroller in driving a stepper motor (a) with low current output, and (b) with high current output. (Schematics adapted by permission from Oriental Motor USA Corp.)

microprocessor only provides the stream of pulses, which are always the same, and not the sequences. A second input pulse to the driver determines the direction of motion. Consequently, only 2 output ports are needed to drive a stepper motor for any displacement, at either direction, at any speed. Therefore, using stepper drivers, the same number of output ports as before (four outputs) can drive twice as many stepper motors. Most drivers provide a choice of full or half stepping by making one of their pins high or low. In this case, three output lines will be necessary. Drivers are very easy to use, are very inexpensive, and simplify programming of the processors. Relatively inexpensive microstepping stepper driver IC chips are also commercially available and may be used for limited microstepping.

Drivers, like microprocessors, may be low current or high current. If they are low current, it is necessary to use power transistors as switches, where the driver turns the transistors on and off, thus driving the stepper motor. If the driver is high current (such as Allegro MicroSystems Inc. A3982 with a current rating of 2 amps), the output port of the driver may directly be connected to the stepper motor, simplifying the circuit. Figure 7.42 is a schematic of a typical stepper motor driver (Allegro Micro Systems A3982 chip). Figure 7.43 shows how it can be used. As shown, the microprocessor provides a pulse train to pin 16 of the driver that automatically sequences the stepper motor for each pulse. The faster the pulses, the faster the stepper motor's rate of rotation. To control the direction of rotation, pin 20 of the driver must change from high to low or from low to high.

Another way to run a stepper motor with a driver is to provide the pulse train to the driver by means other than a microprocessor. As an example, the timer circuit in Figure 7.44 can provide a steady stream of pulses to the driver, causing it to run the

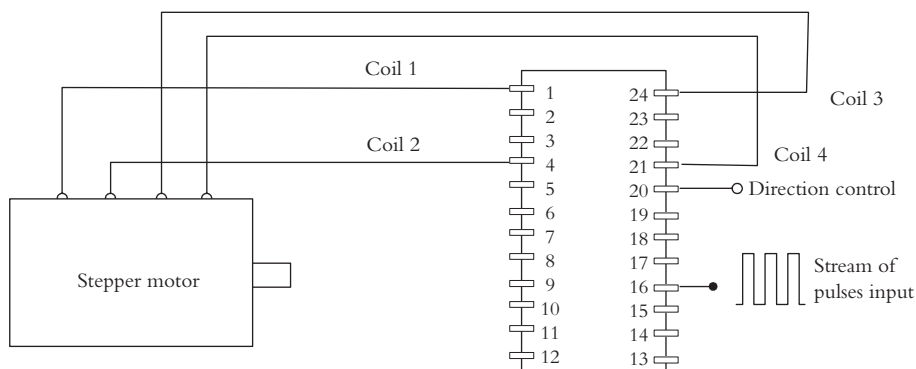


Figure 7.42 Schematic of a typical stepper motor driver.

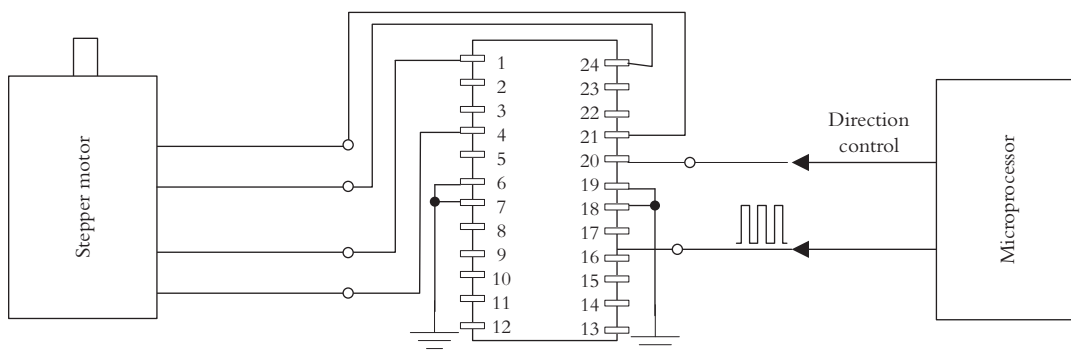


Figure 7.43 Schematic drawing of the application of a driver to run a stepper motor with a microprocessor.

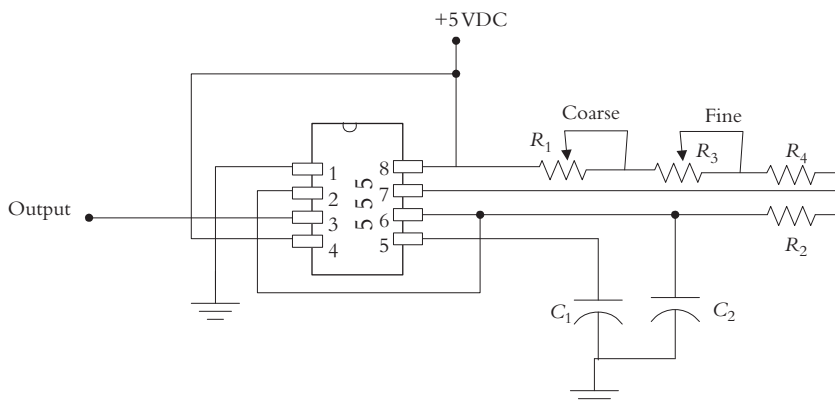


Figure 7.44 Schematic drawing of a timer circuit that creates a pulse train, which can be used to drive a stepper driver. Adjusting the potentiometers will change the period of the output pulses.